Bridging Gaps with Tensor Networks: From the Haldane Gap and Beyond

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Proofs generated by large language models are often difficult to understand, generalize, or formalize directly. Here we verify, digest, and formalize OpenAI's computer-assisted proof of a uniform gap in the periodic spin-1 Heisenberg chain using tensor-network diagrams and Lean 4. Reading the partition-function network along space and imaginary time shows how finite-ring bounds propagate across doubled lengths and inverse temperatures to all longer even rings, sharpening the spin-1 bound from the same finite inputs by a factor of 19 to $\Delta_L > 0.0729$ for all even $L \geq 36$. We identify reflection positivity as the general condition behind the proof and certify uniform gaps in seven additional spin chains and ladders. More broadly, digesting machine-generated proofs into transparent physical mechanisms yields sharper bounds, broader theorems, and reusable formal verifications.

Haldane mapped one-dimensional Heisenberg antiferromagnets onto the nonlinear σ model to predict a gap for integer spins, in contrast to half-integer chains [1–3] (near 0.4105 for spin one [4]). Because the Heisenberg chain is not frustration free, finite-size gap criteria for frustration-free models [5–8] and ground-energy relaxations [9] do not suffice.

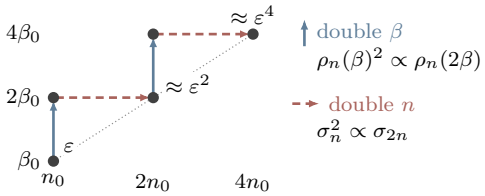
OpenAI recently gave a computer-assisted proof of a unique ground state and a uniform gap $\Delta_L > \frac{\log 20}{784} \simeq 0.003821$ for all even periodic lengths $L \geq 2304$ (and $\Delta_L \gtrsim 4.79 \times 10^{-4}$ for $L \geq 60$) from certified finite partition-function bounds [10]. After verifying those finite inputs in exact arithmetic, we recast the proof in tensor-network diagrams, isolating reflection positivity as the condition behind its spatial inequalities. We then certify uniform gaps for seven additional spin chains and ladders, sharpen the periodic spin-1 bound from the same inputs to $\Delta_L > 0.0729$ for all even $L \geq 36$, and formalize the general theorem in Lean 4 [11].

Thermal purity and the gap.— Consider $H_L = \sum_{j=1}^{L} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}$ and $Z_L(\beta) = \operatorname{Tr} e^{-\beta H_L}$ with periodic indices, spin one, even $L \geq 4$, and unit coupling. Ordering the eigenvalues of H_L with multiplicity defines the gap $\Delta_L = E_1(L) - E_0(L)$ (). For $\beta > 0$, the Gibbs state $\rho_L = e^{-\beta H_L}/Z_L(\beta)$ has ordered weights $w_0 \geq w_1 \geq \cdots$. Its thermal purity $P_\tau(L,\beta) = \operatorname{Tr} \rho_L^2 \leq w_0$ gives ✓L

$$e^{-\beta\Delta_L} = \frac{w_1}{w_0} \leq \frac{1-w_0}{w_0} \leq \frac{1-P_\tau(L,\beta)}{P_\tau(L,\beta)}. \tag{1}$$

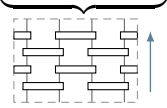
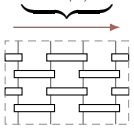
When $P_\tau(L,\beta)$ is close to one (), $P_\tau > 1/2$ forces $w_0 > 1/2$ and hence a unique ground state, while $0 < 1 - P_\tau \leq q < 1/2$ gives $\Delta_L \geq \frac{1}{\beta} \log \frac{1-q}{q}$.

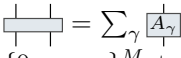
Proof strategy.— Certifying a uniform gap from Eq. (1) requires bounding $P_\tau(L,\beta)$ for all even $L \geq n_0$ at $\beta \propto L$, whereas direct calculation reaches only a finite seed (n_0, β_0). Because H_n is Hermitian, the Gibbs state $\rho_n(\beta)$ is a density operator whose square is proportional to $\rho_n(2\beta)$ and squares the defect $1 - P_\tau$. OpenAI's proof pairs $\rho_n(\beta)$ with a spatial density operator $\sigma_n(\beta)$ whose square is proportional to $\sigma_{2n}(\beta)$ and doubles the ring length $n \mapsto 2n$. From (n_0, β_0), the two squarings alternate along the staircase:

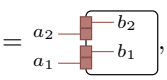


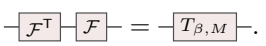
Each step squares the defect $\varepsilon_j = 1 - P$ while doubling both n_j and β_j, so $\frac{\log(1/\varepsilon_j)}{\beta_j}$ stays bounded below. The proof therefore needs three ingredients: a positive semidefinite spatial transfer operator to define the density operator σ_n; a coupling between the spatial $(n \mapsto 2n)$ and thermal $(\beta \mapsto 2\beta)$ squarings; and an interpolation from the dyadic lengths $n_j = 2^j n_0$ to every even $L \in [n_j, 2n_j]$.

Spatial transfer matrix.— Set $\tau = \beta/M$ and let $S_L(\tau)$ be the product of $I - \tau h_{j,j+1}$ over even bonds followed by odd bonds, where $h = \boldsymbol{S} \cdot \boldsymbol{S}$. Contracting this brickwork network in imaginary time or space gives [12]

$$\underbrace{\operatorname{Tr} S_L(\beta/M)^M}_{\text{  }} = Z_L^{(M)}(\beta) = \underbrace{\operatorname{Tr} T_{\beta,M}^{L/2}}_{\text{  }}. \tag{2}$$

In the Cartesian basis $|x\rangle, |y\rangle, |z\rangle$, the generators $G_\alpha = iS^\alpha$ are real antisymmetric matrices. Hence $-h = \sum_{\alpha=x,y,z} G_\alpha \otimes G_\alpha$ and $I - \tau h = \sum_{\gamma=0,x,y,z} A_\gamma \otimes A_\gamma$, where $A_0 = I$ and $A_\alpha = \sqrt{\tau}\, G_\alpha$; diagrammatically,  . Assign bond-label sequences $a, b \in \{0, x, y, z\}^M$ to the odd and even bonds adjacent to a site. Tracing that site through the M slices, with A_{a_t} entering from the odd bond and A_{b_t} leaving on the even bond, defines a real matrix $\mathcal{F}$ independent of L ✓L:

$$\mathcal{F}_{ba} = \operatorname{Tr}[(A_{b_1} A_{a_1}) \cdots (A_{b_M} A_{a_M})] = \text{  },$$

$$T_{\beta,M} = \mathcal{F}^{\mathsf{T}}\mathcal{F} \geq 0, \qquad \text{  }. \tag{3}$$

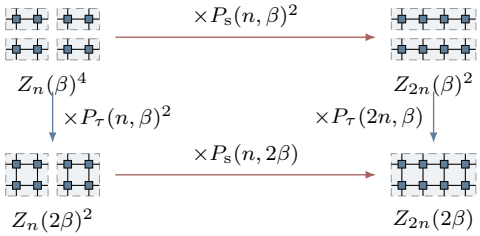
Because each bond gate $I - \tau h$ splits into matching real matrices A_γ, the two sites of an even bond mirror each other across b, contracting $\mathcal{F}$ with its transpose into the positive semidefinite Gram matrix $T_{\beta,M} = \mathcal{F}^{\mathsf{T}}\mathcal{F} \geq 0$—finite-slice reflection positivity [13–15]. For even $n \geq 4$, $\sigma_n^{(M)} = T_{\beta,M}^{n/2}/Z_n^{(M)}(\beta)$ is a density matrix with $(\sigma_n^{(M)})^2 \propto \sigma_{2n}^{(M)}$; taking $M \to \infty$ at fixed (L, β) passes its eigenvalue inequalities to $Z_L(\beta)$ before doubling begins (Sec. SI of Ref. [11]).

Coupling and squaring.— With $Z_n(\beta)$ drawn as the $n \times \beta$ torus , the spatial purity doubles its width and the thermal purity doubles its height:

$$P_{\mathrm{s}}(n,\beta) = \lim_{M\to\infty} \mathrm{Tr}[(\sigma_n^{(M)})^2] = \frac{Z_{2n}(\beta)}{Z_n(\beta)^2} = \frac{\text{  }}{\text{  }^2},$$

$$P_\tau(n,\beta) = \mathrm{Tr}\,\rho_n(\beta)^2 = \frac{Z_n(2\beta)}{Z_n(\beta)^2} = \frac{\text{  }}{\text{  }^2}.$$

Assembling the $2n \times 2\beta$ torus  from four copies of in either order,



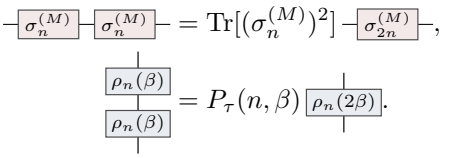
gives the exact coupling identity ✓L

$$P_{\mathrm{s}}(n,2\beta)P_\tau(n,\beta)^2 = P_\tau(2n,\beta)P_{\mathrm{s}}(n,\beta)^2. \tag{4}$$

For any density operator with eigenvalues u_i and purity $P = \sum_i u_i^2$, squaring and renormalizing concentrates the weights ( ↦), giving a new purity P' that obeys ✓L

$$1 - P' = \frac{2\sum_{i<j}(u_i u_j)^2}{P^2} \leq \frac{(1-P)^2}{2P^2} = f(1-P), \tag{5}$$

where $f(z) = z^2/[2(1-z)^2]$. This squaring map acts identically along space and imaginary time:



To advance the bounds $1 - P_{\mathrm{s}}(n,\beta) \leq p$ and $1 - P_\tau(2n,\beta) \leq q$ from (n,β) to $(2n,2\beta)$, we alternate torus gluing with squaring along the two directions:

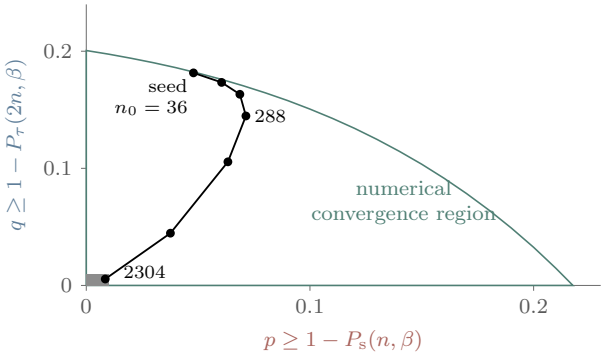
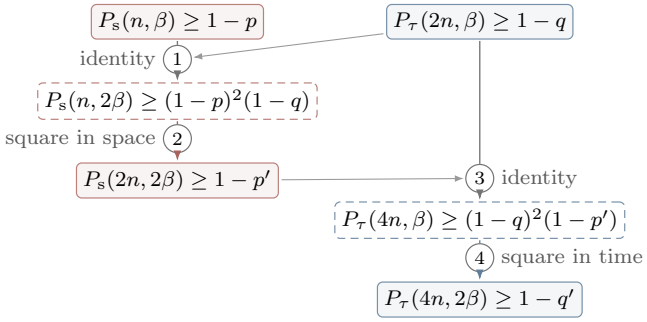


FIG. 1. Convergence region of the doubling map (6) in the defect bounds $p \geq 1 - P_{\mathrm{s}}(n,\beta)$, $q \geq 1 - P_\tau(2n,\beta)$. A seed (p_0, q_0) inside the teal boundary gives a uniform gap: doubling $(n,\beta) \mapsto (2n,2\beta)$ drives (p_j, q_j) to $(0,0)$, and inside the gray square $[0, 0.01]^2$ the bound $\varepsilon_{j+1} = 5\varepsilon_j^2$ takes over. Markers: the spin-1 trajectory from $n_0 = 36$; the small p_0 pulls q_j down over three steps ($n = 288$) and reaches the square at step six ($n = 2304$).



Dropping a purity factor ≤ 1 in Eq. (4) (moves 1, 3) and applying Eq. (5) (moves 2, 4) gives ✓L

$$\begin{aligned} p' &= f(1 - (1-p)^2(1-q)), \\ q' &= f(1 - (1-q)^2(1-p')), \end{aligned} \tag{6}$$

which maps bounds (p, q) at (n,β) to (p', q') at $(2n, 2\beta)$ inside the convergent region of Fig. 1.

From the finite seed to the uniform gap.— Once (p_j, q_j) enters $[0, 0.01]^2$ (gray in Fig. 1), $f(1 - (1-\varepsilon)^3) \leq 5\varepsilon^2$ drives all further doubling steps analytically (Sec. SI.F of Ref. [11]). We start from OpenAI's certified seed:

Lemma 1 (Finite seed [10] ✓L)**.** *At $\beta_0 = 49/4$, the periodic spin-1 chain satisfies $P_{\mathrm{s}}(36, \beta_0) \geq 1 - 47916/10^6$ and $P_\tau(72, \beta_0) \geq 1 - 181521/10^6$.*

Although $q_0 = 0.181521$ exceeds the symmetric fixed point $\varepsilon_* \simeq 0.1273$, the smaller p_0 pulls q' below q, and six steps of Eq. (6) reach $(p_6, q_6) \leq (0.0085, 0.0056)$ at $(n_6, \beta_6) = (2304, 784)$ (Fig. 1; Sec. SI.E of Ref. [11]).

Restarting at $(2304, 784)$ with $\varepsilon_0 = 0.01$ and $\varepsilon_{j+1} = 5\varepsilon_j^2$ gives $\varepsilon_j = 20^{-2^j}/5$ at $n_j = 2304\,2^j$ and $\beta_j = 784\,2^j$. To cover every even $L \geq 2304$, choose j with $n_j \leq L \leq 2n_j$. Because the eigenvalues of $\sigma_u^{(M)}$ lie in $[0,1]$ and sum to 1, raising them to a power $v/u \geq 1$ lowers their sum

(↦), giving $Z_v(\beta) \le Z_u(\beta)^{v/u}$ for even $v \ge u$ as $M \to \infty$ and bounding $P_\tau(L,\beta_j)$ ( ): ✓L

$$P_\tau(L,\beta_j) \ge [P_\tau(2n_j,\beta_j)P_{\rm s}(n_j,\beta_j)^2]^{L/(2n_j)} \ge (1-\varepsilon_j)^3 \ge 1-3\varepsilon_j. \quad (7)$$

With $\varepsilon_j \le 0.01$, $3\varepsilon_j/(1-3\varepsilon_j) < 20^{-2^j}$, so Eq. (1) proves a unique ground state and ✓L

$$\Delta_L > \frac{\log 20}{784} \simeq 0.003821 \qquad \text{for all even } L \ge 2304.$$

Bounding the finite seed.— Certifying a seed at (n_0,β_0) reduces to bounding the numerators of

$$P_{\rm s}(n_0,\beta_0) = \frac{Z_{2n_0}(\beta_0)}{Z_{n_0}(\beta_0)^2}, \qquad P_\tau(2n_0,\beta_0) = \frac{Z_{2n_0}(2\beta_0)}{Z_{2n_0}(\beta_0)^2} \quad (8)$$

from below and their denominators from above. When $2n_0$ is too large for direct diagonalization ($3^{72} \simeq 2.3 \times 10^{34}$ at $n_0 = 36$), Ref. [10] bounds these four partition functions from short-ring traces on $4 \le n \le 12$ sites ($3^n \le 5.3\times10^5$) and a periodic matrix-product trial state at length $N = 2n_0$:

	Quantity	Bound	Method
	$Z_{2n_0}(\beta_0)$	lower	MPS at $2n_0$ + short-ring traces
	$Z_{n_0}(\beta_0)$	upper	MPS at $2n_0$ + short-ring traces
	$Z_{2n_0}(2\beta_0)$	lower	MPS at $2n_0$
	$Z_{2n_0}(\beta_0)$	upper	short-ring traces

Five steps turn these inputs into Lemma 1 at $n_0 = 36$ and $\beta_0 = 49/4$ (Sec. SII of Ref. [11]).

Step 1: One-site spatial transfer operator. Unlike the two-site Gram matrix $T_{\beta,M}$ (even lengths only), the continuous-time self-adjoint one-site transfer operator K_β of Ref. [10] gives $Z_n(\beta)$ for all $n \ge 4$:

$$Z_n(\beta) = \operatorname{Tr} K_\beta^n = \quad , \qquad n \ge 4.$$

Shifting the energy per site by $\kappa = 700741/500000$ (so the trial ground-state energy in Step 3 is negative) defines $\widehat{K} = e^{-\beta_0\kappa}K_{\beta_0}$ with real eigenvalues λ_i of either sign and shifted traces

$$\widehat{Z}_n(\beta) = e^{-n\beta\kappa}Z_n(\beta), \qquad \widehat{Z}_n(\beta_0) = \sum_i \lambda_i^n, \quad (9)$$

which leave both purities unchanged and sample the same $\{\lambda_i\}$ at every $n \ge 4$.

Step 2: Symmetry sectors from twisted rings. To tighten the subleading sum at n_0, the commuting π spin rotations split $\widehat{K}$ into a rotation-invariant sector J and three degenerate sectors N permuted by axis rotations. Twisting the closing bond by a π rotation P negates two of the three N sectors, giving

$$\widehat{Z}_n = J_n + 3N_n,$$
$$\widehat{Z}_n(P) = J_n - N_n, \quad (10)$$

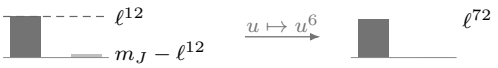
with $J_n = \sum_{i\in J}\lambda_i^n$ and $N_n = \sum_{i\in N}\lambda_i^n$ at β_0. Integer polynomial approximations to $e^{-\beta_0 H_n/2}$ enclose both traces for $4 \le n \le 12$; solving Eq. (10) at $n = 12$ (which divides n_0) gives the reference twelfth-moment bounds $J_{12} \le m_J = 1.0657205$ and $N_{12} \le m_N = 0.2783155$.

Step 3: Partition-function lower bound from a trial state. On a ring of length $N = 2n_0$, integer tensors A (bond dimension 26) define a periodic MPS ψ whose one-bond energy contracts as an N-step matrix trace,

$$\mathcal{E} = \langle\psi, h_{1,2}\psi\rangle = \quad ,$$

and whose norm $\mathcal{N} = \langle\psi,\psi\rangle$ omits the h box. Translation invariance gives $\langle\psi, H_N\psi\rangle/(N\mathcal{N}) = \mathcal{E}/\mathcal{N} < -\kappa$, so $E_0(N) < -N\kappa$ and retaining only the ground-state weight gives $\widehat{Z}_N(\beta) \ge e^{-\beta(E_0(N)+N\kappa)} > 1$ for all $\beta > 0$, supplying both $\sum_i \lambda_i^{2n_0} > 1$ at β_0 and $\widehat{Z}_{2n_0}(2\beta_0) > 1$.

Step 4: Lower bound on the leading spatial eigenvalue. To bound $\widehat{Z}_{2n_0}(\beta_0) \ge \lambda_0^{2n_0}$ from below, we show that J contains an eigenvalue with $|\lambda_0| > \ell$ ($\ell = 0.999$ at $n_0 = 36$). Write $u_i = \lambda_i^{12} \ge 0$, so $\lambda_i^{2n_0} = u_i^{2n_0/12}$. If every $|\lambda_i| \le \ell$, then $u_i \le \ell^{12}$; since $u \mapsto u^{2n_0/12}$ is convex and $\ell^{12} < m_J < 2\ell^{12}$, $\sum_{i\in J} u_i^{2n_0/12}$ at mass $\le m_J$ is maximized by one entry ℓ^{12} and remainder $m_J - \ell^{12}$:



Raising to the power $2n_0/12$ suppresses the remainder and bounds each N sector by $m_N^{2n_0/12}$, giving at $2n_0 = 72$ (< 0.932)

$$\widehat{Z}_{2n_0}(\beta_0) \le \ell^{2n_0} + (m_J - \ell^{12})^{2n_0/12} + 3m_N^{2n_0/12} < 1,$$

contradicting $\widehat{Z}_{2n_0}(\beta_0) > 1$ from Step 3; since $m_N < \ell^{12}$, the eigenvalue $|\lambda_0| > \ell$ lies in J.

Step 5: Polynomial upper bounds on the remaining spectrum. To bound $\widehat{Z}_{n_0}(\beta_0)$ and $\widehat{Z}_{2n_0}(\beta_0)$ from above, we bound the remaining eigenvalues in each sector $\mathcal{S} \in \{J, N\}$ using a nonnegative polynomial $F(x) = \sum_{j=4}^{12} f_j x^j$:

$$F(\lambda_i) \le \sum_{a\in\mathcal{S}} F(\lambda_a) = \sum_{j=4}^{12} f_j \mathcal{S}_j \le B, \qquad i \in \mathcal{S}.$$

No eigenvalue in $\mathcal{S}$ lies where $F(x) > B$ ( ). Two degree-12 polynomials rule out $|\lambda| \ge U = 1.00139$ in J and $|\lambda| \ge V = 0.872$ in N (Sec. SII.A.c of Ref. [11]).

Removing $\lambda_0^{12} > \ell^{12}$ leaves twelfth moment at most $m_J - \ell^{12}$ in J, while convexity with upper bound V^{12} ($V^{12} < m_N < 2V^{12}$) bounds each N sector by one entry at V^{12} and the remainder $m_N - V^{12}$ ( ). For $k \in \{n_0, 2n_0\}$, the subleading sum $\sum_{i\ne0}\lambda_i^k$ is at most

$$R_k = (m_J - \ell^{12})^{k/12} + 3\big[V^k + (m_N - V^{12})^{k/12}\big].$$

TABLE I. Certified uniform gap bounds for reflection-positive spin chains and ladders, covering every even length L (L counts sites, two-spin dimer cells, or two-spin rungs; $\boldsymbol{S}^2 = S(S+1)I$, σ^α Pauli matrices). "Gap $\geq$" gives the certified lower bound on $\Delta_L = E_1(L) - E_0(L)$ (the interval bound of the certificate) and links to the Lean 4 theorem that proves the rational-chain bound from the seed pair as hypothesis, equal in the quoted digits except where the chain was rounded more coarsely; the single-ion row's Lean theorem starts from the direct seed $(6, 9/4)$ and proves 0.6946; "Numerical" lists uncertified 12–20-spin estimates (or bulk DMRG [4] for $S = 1$ Heisenberg). The mixed-field Ising bound is verified unconditionally in the Lean kernel.

Model	Hamiltonian	Even lengths	Gap $\geq$	Numerical
Staggered field, $S = \frac{1}{2}$	$\sum_j \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1} - \sum_j (-1)^j S^z_j$	$L \geq 8$ sites	1.1489 ✓L	1.874
Dimerized, $S = \frac{1}{2}$	$\sum_j [1 + (-1)^j/2]\, \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}$	$L \geq 4$ cells	0.5347 ✓L	1.189
Dimerized, $S = 1$	$\sum_j [1 + 9(-1)^j/10]\, \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}$	$L \geq 2$ cells	0.5288 ✓L	1.763
Single-ion anisotropy, $S = 1$	$\sum_j \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1} + 4\sum_j (S^z_j)^2$	$L \geq 12$ sites	0.8994 ✓L	2.269
Heisenberg ladder, $S = \frac{1}{2}$	$\sum_{j,a=1,2} \boldsymbol{S}_{j,a} \cdot \boldsymbol{S}_{j+1,a} + 4\sum_j \boldsymbol{S}_{j,1} \cdot \boldsymbol{S}_{j,2}$	$L \geq 4$ rungs	1.1024 ✓L	3.138
Ising ladder	$-\sum_{j,a} \sigma^z_{j,a}\sigma^z_{j+1,a} - \sum_j \sigma^z_{j,1}\sigma^z_{j,2} - 3\sum_{j,a} \sigma^x_{j,a}$	$L \geq 4$ rungs	0.6368 ✓L	2.662
Mixed-field Ising	$-\sum_j \sigma^z_j\sigma^z_{j+1} - \frac{1}{4}\sum_j \sigma^x_j - \frac{1}{20}\sum_j \sigma^z_j$	$L \geq 6$ sites	0.3922 ✓L	1.190
Heisenberg, $S = 1$	$\sum_j \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}$	$L \geq 36$ sites	0.0729 ✓L	0.410

Bounding each numerator of Eq. (8) below by its leading term and each denominator above by its leading term ($|\lambda_0|^{n_0} > \ell^{n_0}$ and U^{2n_0}) plus R_k gives ✓L

$$P_{\mathrm{s}}(n_0, \beta_0) \geq \left[1 + R_{n_0}/\ell^{n_0}\right]^{-2},$$
$$P_\tau(2n_0, \beta_0) \geq \left[U^{2n_0} + R_{2n_0}\right]^{-2};$$

exact rational evaluation at $(36, 72)$ returns the two bounds of Lemma 1.

Because Boltzmann weights $z_i > 0$ satisfy $\sum_i z_i^r \leq (\sum_i z_i)^r$ for $r > 1$, short-ring trace bounds at β_0 also bound the moments at $r\beta_0$. Whereas Ref. [10] lifted a separate $n \leq 10$ table at $\beta_0' = 21/2$ by $r = 5/2$ to get $\Delta_L \gtrsim 4.79 \times 10^{-4}$ for $L \geq 60$, lifting the stronger $\beta_0 = 49/4$ ($n \leq 12$) table by $r = 17/8$ with a 120-site trial state and tighter polynomial bounds U, V certifies $(p_0, q_0) = (0.0306, 0.0480)$ at $n_0 = 60$Ten steps of Eq. (6) and bridging $36 \leq L \leq 72$ from $(36, \beta_0)$ then prove $\Delta_L > 0.0729$ for all even $L \geq 36$ ✓L (Sec. SII.B of Ref. [11]).

Reflection-positive models.— The spatial Gram factorization (3) relies on locality and bond reflection positivity [14, 15], not on SU(2) symmetry. Consider a periodic ring of $L \geq 2$ unit cells with Hamiltonian $H_L = \sum_{j=1}^{L} h_{j,j+1} + \sum_{j=1}^{L} o_j$ ($j + L \equiv j$), where h couples neighboring cells and o acts within a cell. Suppose in some local spin basis there is a real symmetric involution F ($F^2 = I$, such as dimer spin exchange) for which o is real symmetric with $FoF = o$ and

$$-h = \sum_\alpha C_\alpha \otimes FC_\alpha F,$$



with each C_α real symmetric or antisymmetric and the mirror (dashed) between cells. Conjugating odd cells by F gives both sublattices the same trace matrix $\mathcal{F}$, so the Gram form (3) and the doubling recurrence apply unchanged (Sec. SI.A of Ref. [11]).

A convergent seed at even n_0 then certifies a unique ground state and a uniform gap for all even $L \geq n_0$ by applying Eq. (7) on each dyadic interval $[2^j n_0, 2^{j+1} n_0]$ ✓L. Table I applies this theorem to staggered-field chains [16], bond-alternating chains [17, 18], chains with single-ion anisotropy [19], Heisenberg ladders [20], and Ising models, with finite seeds certified from verified short-ring spectra or exact matrix arithmetic (Sec. SIII of Ref. [11]).

Formal verification.— References [10, 21] were not accompanied by a Lean proof, and formalizing their continuous-time kernel directly would require path-space trace identities not yet in Mathlib. Our finite Gram matrix (3) and fixed-(L, β) limit (2) instead reduce the spatial inequalities to finite-dimensional matrix algebra ✓L and yield a general reflection-positive gap theorem in Lean 4 [22–24] (Sec. SIV of Ref. [11]; each ✓L links to a Lean theorem).

A computer-assisted proof also relies on finite seed computations that are too large for direct diagonalization inside a proof-assistant kernel. Separating the universal doubling theorem from the finite seed lets the formalization work at two levels: taking certified seed bounds as hypotheses instantiates all eight models in Table I, whereas reducing the seed to exact integer identities lets the Lean kernel verify the finite arithmetic itself—proving the mixed-field Ising gap unconditionally ✓L (Sec. SIII.C of Ref. [11]) and checking the spin-1 trial states and shortest ring traces so that only the remaining short-ring trace bounds enter as hypotheses ✓L.

Discussion.— Beyond periodic rings, the same diagrammatic method extends to open spin-1 chains with endpoint magnetic fields [21]: replacing the spatial trace by boundary vectors and tracking their overlap (∠) with the leading spatial eigenvector gives a diagrammatic proof and Lean formalization that sharpen the gap bound to $\Delta_n > 0.01556$ for all odd $n \geq 1921$ ✓L (Sec. SV of Ref. [11]). More broadly, as machine-assisted proofs be-

come more common, raw proofs are often hard to understand, generalize, or formalize directly; verifying the finite inputs first, digesting the argument into a transparent mechanism, and only then formalizing the general theorem turns a raw proof into sharper bounds, new certified models, and a reusable formal library.

In two dimensions, a spacetime cube identity couples two spatial purities with the thermal purity when both lattice directions are reflection positive [23]; for the honeycomb AKLT model [25–27], the spin-3/2 bond projector violates standard reflection positivity, leaving spatial gluing open.

Finally, uniform gap bounds supply the hypothesis of correlation-decay and entanglement area-law theorems in one and two dimensions [28–32], whose proofs have also been formalized in Lean [33, 34]. Checking the remaining spin-1 short-ring trace recursions in the kernel motivates verified numerical linear algebra in the proof assistant.

We sincerely thank Xiao-Liang Qi for insightful discussions and hospitality at Stanford University, and thank Ignacio Cirac, Shengqi Sang, Yimu Bao, and Junkai Dong for helpful discussions. Part of this work was also carried out during a visit to the Kavli Institute for Theoretical Physics (KITP), supported in part by grant NSF PHY-2309135; we thank KITP for its hospitality. This research was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy—EXC-2111—390814868. Following the Agentic Publication Protocol [35], the code, gap certificates, Lean 4 formalization [23], and proof blueprint [24] are available in the companion repository and interactive project website [36] (Sec. SIV of Ref. [11]). We welcome open-science contributions of new or improved gap certificates and Lean 4 formalizations [36].

Note added.—An initial version of this work was circulated at Stanford on October 7, 2026, and presented at the KITP *Intermediate-Scale AI for Physics* conference [37]. We note the independent expository account by Tasaki [38] and the independent preprint by Grover [39], which appeared after this work was completed.

AI-use declaration.—S. L. prompted GPT-6-Astra (OpenAI) to recast the proofs of Refs. [10, 21] in tensor-network diagrams, and used TeXRA [40], a multi-agent autoformalization workflow [33, 34], and OpenAI Codex with GPT-6-Astra and GPT-6-Sol for subsequent calculations, formalization, and manuscript preparation. All mathematical arguments, numerical certificates, and formal proofs were verified by the author.

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Supplemental Material

This Supplemental Material contains the full proofs, seed certificates, and Lean 4 formalization details. Section SI constructs the finite-dimensional spatial transfer matrix $T_{\beta,M}$ for any reflection-positive ring Hamiltonian, takes the continuum limit $M \to \infty$, and proves the periodic scale-doubling theorem: a finite seed (n_0, β_0) with both defects at most $1/100$ gives a uniform gap on all even lengths $L \geq n_0$, applied to the spin-1 chain from $(2304, 784)$. Section SII proves the periodic spin-1 Heisenberg seeds from the one-site Poisson transfer operator K_β, symmetry-sector trace recursions, polynomial spectral filters, and matrix-product trial states, first at $(n_0, \beta_0) = (36, 12)$ and then at $\beta_0 = 49/4$ to obtain $\Delta_L > 0.07290$ for all even $L \geq 36$. Section SIII gives the reflection-positive factorizations, finite seeds, and gap bounds for the seven other spin chains and ladders in Table I, including the unconditional Lean-kernel proof for the mixed-field Ising chain. Section SIV describes the Lean 4 formalization of the general criterion, the model theorems, and the kernel-checked seed arithmetic. Section SV extends the diagrammatic doubling proof and Lean 4 formalization to open spin-1 Heisenberg chains with endpoint fields, proving $\Delta_n > 0.01556$ for all odd $n \geq 1921$.

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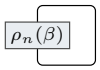
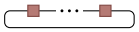
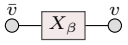
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Graphical notation

Every diagram below is a tensor network built from the elements in the following legend. Red marks space, blue marks imaginary time, teal marks certified conclusions, grays mark spectra, weights, and boundary vectors, and ochre marks regions ruled out.

	physical leg: one site, through imaginary time		closed physical loop: the trace Tr
	virtual leg: the bond between two sites		closed virtual loop: the trace Tr_{v}
	joined legs are contracted		boundary vectors on an open chain
	local MPO tensor (thermal, blue)		transfer vertex: W_β with physical legs traced
	Gibbs state on n sites (thermal, blue)		spatial transfer operator ($T_{\beta,M}, K_\beta, X_\beta$, red)
	$Z_n(\beta)$: a torus of width n and height β		length doubled, inverse temperature doubled

SI: Finite Trotter slices and the periodic doubling proof

SI.A. Finite Trotter slices and the two-site transfer matrix

OpenAI's proof uses a continuous-time spatial integral kernel [10], whose trace identities require infinite-dimensional path-space integration. To formalize the general gap theorem in Lean with finite-dimensional linear algebra only, we discretize imaginary time into M brickwork slices. The spatial transfer operator is then a finite Gram matrix $T_{\beta,M} = \mathcal{F}^{\mathsf{T}}\mathcal{F} \succeq 0$, and a fixed-$(L,\beta)$ matrix product limit passes its moment inequalities to the exact partition functions $Z_L(\beta)$.

Consider the periodic spin-1 Heisenberg Hamiltonian

$$H_L = \sum_{j=0}^{L-1} h_{j,j+1}, \qquad h = \sum_{\alpha=x,y,z} S^\alpha \otimes S^\alpha, \qquad Z_L(\beta) = \mathrm{Tr}\, e^{-\beta H_L}, \tag{S1}$$

with indices taken modulo L, exchange coupling equal to one, and $\beta > 0$. In the Cartesian basis $s, s' \in \{x, y, z\}$, set

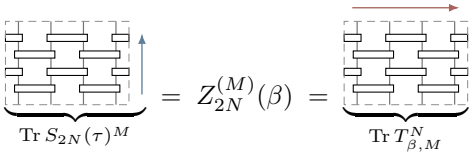
$$(S^\alpha)_{ss'} = -i\epsilon_{\alpha ss'}, \qquad G_\alpha = iS^\alpha, \qquad -h = \sum_\alpha G_\alpha \otimes G_\alpha. \tag{S2}$$

Here $\epsilon_{\alpha ss'}$ is the Levi-Civita symbol, Tr denotes the matrix trace, and I denotes the identity on the relevant space. Each G_α is real antisymmetric and has operator norm one.

For $L = 2N$, $N \geq 1$, and a positive integer M, let $\tau = \beta/M$. Define the linearized brickwork slice and its trace by

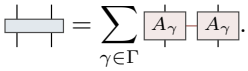
$$\begin{aligned} S_{2N}(\tau) &= \prod_{k=0}^{N-1} (I - \tau h_{2k,2k+1}) \prod_{k=0}^{N-1} (I - \tau h_{2k+1,2k+2}), \\ Z_{2N}^{(M)}(\beta) &= \mathrm{Tr}\, S_{2N}(\beta/M)^M. \end{aligned} \tag{S3}$$

Products run from left to right in increasing index, and the trace is real because every factor is. Equation (S3) multiplies M slices; the same network read along space multiplies two-site columns,

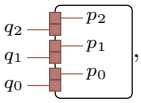


(S4)

The moment inequalities of Sec. SI.B need $Z_{2N}^{(M)}(\beta) = \operatorname{Tr} T_{\beta,M}^{N}$ with a positive semidefinite column. A one-site column, like the continuous-time thermal-MPO kernel of Ref. [41], is real symmetric but need not be positive; two sites mirrored across a bond give a Gram matrix instead. Let $\Gamma = \{0, x, y, z\}$, with $A_0 = I$ and $A_\alpha = \sqrt{\tau}\, G_\alpha$. Each bond gate $I - \tau h$ splits into matching matrices A_γ on its two sites,

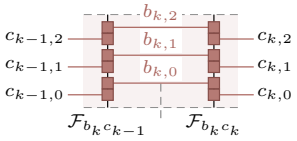
$$I - \tau h = \sum_{\gamma\in\Gamma} A_\gamma \otimes A_\gamma, \qquad$$  $$= \sum_{\gamma\in\Gamma}$$. (S5)

An imaginary-time bond sequence $p \in \mathcal{D}_M = \Gamma^M$ lists one label per slice. Define the single-site trace matrix, independent of N, by

$$\mathcal{F}_{pq} = \operatorname{Tr}\left[\prod_{t=0}^{M-1} (A_{p_t} A_{q_t})\right] =$$  , $\qquad p, q \in \mathcal{D}_M,$ (S6)

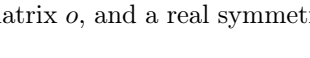
drawn for $M = 3$ with A_{q_t} entering from the odd bond on the left and A_{p_t} leaving on the even bond to the right.

To contract the network in space, assign sequences b_k to the even bonds and c_k to the odd bonds. Across the mirror of an even bond the left site is the reflected right site, so the column is $\mathcal{F}$ contracted with its own transpose:



$$\sum_{b_k\in\mathcal{D}_M} \mathcal{F}_{b_k c_{k-1}} \mathcal{F}_{b_k c_k} = (\mathcal{F}^{\mathsf{T}}\mathcal{F})_{c_{k-1} c_k}.$$ (S7)

Summing all bond labels and tracing every site gives ✓L

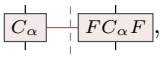
$$\begin{aligned} Z_{2N}^{(M)}(\beta) &= \sum_{\substack{b_0,\dots,b_{N-1}\in\mathcal{D}_M \\ c_0,\dots,c_{N-1}\in\mathcal{D}_M}} \prod_{k=0}^{N-1} \mathcal{F}_{b_k c_{k-1}} \mathcal{F}_{b_k c_k} \\ &= \sum_{c_0,\dots,c_{N-1}\in\mathcal{D}_M} \prod_{k=0}^{N-1} (\mathcal{F}^{\mathsf{T}}\mathcal{F})_{c_{k-1} c_k} = \operatorname{Tr}(\mathcal{F}^{\mathsf{T}}\mathcal{F})^N = \end{aligned}$$  , (S8)

where $c_{-1} = c_{N-1}$: the middle equality is Eq. (S7) at each even bond, and the last closes the N columns into a ring.

Thus the spatial transfer matrix ✓L

$$T_{\beta,M} = \mathcal{F}^{\mathsf{T}}\mathcal{F} = \mathcal{F}^{\dagger}\mathcal{F} \succeq 0$$ (S9)

advances *two sites*. Because $\mathcal{F}$ is real, $v^{\dagger} T_{\beta,M} v = \|\mathcal{F} v\|^2 \geq 0$ without $\mathcal{F}$ itself being symmetric or positive. The mirror across the even bond in Eq. (S7) is the matching of real factors in Eq. (S5), that is, reflection positivity [13–15].

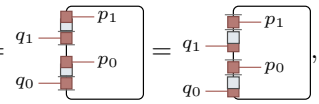
The models of Sec. SIII carry on-site terms and multi-site cells, which Eq. (S6) omits. The same contraction covers any real reflection-positive cell model, with lengths counting cells. Take a finite-dimensional cell space with real symmetric or antisymmetric bond matrices C_α, a real symmetric on-site matrix o, and a real symmetric involution F in the bond mirror  , such that ✓L

$$F^2 = I, \qquad FoF = o, \qquad h = -\sum_{\alpha} C_\alpha \otimes F C_\alpha F.$$ (S10)

The transpose signs cancel between the two factors of each bond term, so $h^{\mathsf{T}} = h$ and $H_L = \sum_j h_{j,j+1} + \sum_j o_j$ is self-adjoint. Insert the on-site layer $\prod_i (I - \tau o_i)$ between the even and odd bond layers and expand with

$$A_0 = I, \qquad A_\alpha = \sqrt{\tau}\, C_\alpha, \qquad O = I - \tau o, \qquad I - \tau h = \sum_{\gamma} A_\gamma \otimes F A_\gamma F.$$

Each even cell now carries the on-site factor O (blue box) and the two reflections F (gray ticks) on its worldline, so Eq. (S6) becomes

$$\mathcal{F}_{pq} = \operatorname{Tr} \prod_{t=0}^{M-1} (A_{p_t} O F A_{q_t} F) =$$  , (S11)

with sequences over the enlarged bond-label alphabet. The last equality is the odd cell, $FA_{p_t}FOA_{q_t}$ ($FO = OF$, $F^2 = I$), so both sublattices give the same $\mathcal{F}$ and Eqs. (S8) and (S9) follow; everything below applies with L counting cells.

SI.B. Spatial and thermal purity inequalities at finite M

The recurrence of Sec. SI.D needs three facts: the spatial purity is at most one, squaring a distribution shrinks its defect quadratically, and power sums interpolate between lengths. Positivity of $T_{\beta,M}$ expresses $Z_L^{(M)}(\beta)$ as an even moment sum of nonnegative weights, which supplies all three at finite M. At fixed M, β, let $\lambda_i \geq 0$ be the eigenvalues of the transfer matrix ( ) and put $\mu_i = \sqrt{\lambda_i}$. For every even $L \geq 2$, Eq. (S8) gives

$$Z_L^{(M)}(\beta) = \sum_i \mu_i^L. \tag{S12}$$

Doubling n replaces the weights μ_i^n by their squares, so we need the defect of a squared family in terms of the original defect. For any finite nonnegative family a_i, let $S_k = \sum_i a_i^k$. Then ✓

$$S_2 \leq S_1^2, \qquad 2(S_2^2 - S_4) \leq (S_1^2 - S_2)^2. \tag{S13}$$

Cut the square of side S_1 by the weights ( ): the dark diagonal blocks have area S_2 and each symmetric pair of light blocks has area $2a_ia_j$, so

$$S_1^2 - S_2 = \sum_{i<j} 2a_ia_j, \qquad 2(S_2^2 - S_4) = \sum_{i<j} (2a_ia_j)^2 \leq \Big(\sum_{i<j} 2a_ia_j\Big)^2,$$

a sum of squares being at most the square of the sum.

If $\sum_i a_i = 1$, the purity is $P = \sum_i a_i^2 > 0$. Squaring and normalizing concentrates the distribution (↦) onto a_i^2/P, of purity $P' = S_4/P^2$. Thus ✓

$$1 - P' \leq f(1 - P), \qquad f(z) = \frac{z^2}{2(1-z)^2}. \tag{S14}$$

The map f is nondecreasing on $0 \leq z < 1$, since $z/(1-z)$ is nonnegative and increasing there. For $0 < z < 1/2$, $f(z) < z$ ( ), and squaring makes a small $1 - P$ quadratically smaller.

Section SI.F must also bound even lengths between dyadic scales, which it does by comparing power sums of different orders. For $0 < u \leq v$, nonnegative weights also satisfy ✓

$$\sum_i \mu_i^v \leq \left(\sum_i \mu_i^u\right)^{v/u}. \tag{S15}$$

If the sum on the right is zero, the claim is immediate. Otherwise normalize μ_i^u to probabilities a_i; the power $v/u \geq 1$ lowers every bar (↦), so $\sum_i a_i^{v/u} \leq \sum_i a_i = 1$.

Substituting Eq. (S12) gives, at fixed β, the closed scalar inequalities ✓

$$\begin{aligned} Z_{2n}^{(M)} &\leq (Z_n^{(M)})^2, && \text{Eq. (S13) with } a_i = \mu_i^n, \\ 2((Z_{2n}^{(M)})^2 - Z_{4n}^{(M)}) &\leq ((Z_n^{(M)})^2 - Z_{2n}^{(M)})^2, && \text{Eq. (S13) with } a_i = \mu_i^n, \\ Z_v^{(M)} &\leq (Z_u^{(M)})^{v/u}, && \text{Eq. (S15)}, \end{aligned} \tag{S16}$$

for even $n \geq 2$ and even $v \geq u \geq 2$ (temperature argument suppressed only here). These three inequalities are all the recurrence uses; Sec. SI.C passes them to the exact $Z_L(\beta)$.

SI.C. Taking the Trotter limit $M \to \infty$

To pass the finite-M inequalities in Eq. (S16) to the exact partition functions, we need the matrix product limit $S_L(\beta/M)^M \to e^{-\beta H_L}$. Each inequality involves finitely many fixed pairs (L, β), so the error may grow with L and a telescoping estimate at fixed L and β suffices. Use the operator norm, which is submultiplicative and satisfies $\|I\| = 1$. For $\|X\|, \|Y\| \le K$, $K \ge 1$, the difference of k-fold products telescopes into k chains that differ in one link, each of norm at most $K^{k-1}\|X - Y\| \le K^k \|X - Y\|$:

$$X^k - Y^k = \sum_{r=0}^{k-1} \underbrace{X \cdots X}_{k-1-r} (X - Y) \underbrace{Y \cdots Y}_{r}, \qquad \left\|X^k - Y^k\right\| \le kK^k \|X - Y\|. \tag{S17}$$

Suppose $M(X_M - I) \to A$, and set $Y_M = e^{A/M}$. Then $M(Y_M - I) \to A$ and $Y_M^M = e^A$. With $C = \|A\| + 1$, convergence implies $\|X_M\|, \|Y_M\| \le 1 + C/M$ for sufficiently large M, and Eq. (S17) gives the quantitative comparison

$$\begin{aligned} \left\|X_M^M - e^A\right\| &\le (1 + C/M)^M \|M(X_M - Y_M)\| \\ &\le e^C \|M(X_M - Y_M)\| \longrightarrow 0. \end{aligned} \tag{S18}$$

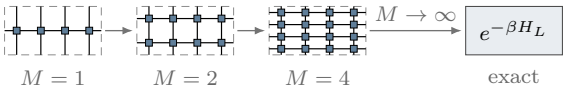
In particular, if $g(0) = I$ and $g'(0) = A$, then $g(1/M)^M \to e^A$.

For fixed L, β, list the negated bond operators in the order of Eq. (S3), including the negated on-site operators when present, as $B_1, \ldots, B_k$. Their sum is $-H_L$, so the polynomial $g(t) = \prod_{i=1}^k (I + \beta t B_i)$ has $g'(0) = -\beta H_L$. Hence ✓L

$$S_L(\beta/M)^M \longrightarrow e^{-\beta H_L}, \qquad Z_L^{(M)}(\beta) \longrightarrow Z_L(\beta). \tag{S19}$$

More explicitly, Taylor expansion gives $g(1/M) - e^{-\beta H_L/M} = O(M^{-2})$. Equation (S18) therefore bounds the operator error by $e^C \kappa/M$ for a finite κ depending on L and β, with $C = \|\beta H_L\| + 1$, once M is large enough. Trace convergence follows from $|\mathrm{Tr}\, A| \le 3^L \|A\|$, with 3^L the dimension of the spin-1 ring.

Taking $M \to \infty$ at each fixed (L, β) occurring in Eq. (S16) passes all three inequalities to the exact partition functions $Z_L(\beta)$ ✓L:



The recurrence therefore starts from exact partition functions, and no Trotter error enters any later step.

SI.D. Torus coupling identities and the asymmetric recurrence

Once the spatial moment inequalities hold for the exact partition functions $Z_L(\beta)$, we couple the spatial and thermal purities on a torus. A torus of width n and height β (▭) represents $Z_n(\beta)$; the spatial purity joins two copies side by side and the thermal purity stacks them in imaginary time:

$$P_s(n, \beta) = \frac{Z_{2n}(\beta)}{Z_n(\beta)^2} = \frac{\square\square}{\square^2}, \qquad P_\tau(L, \beta) = \frac{Z_L(2\beta)}{Z_L(\beta)^2} = \frac{\square}{\square^2}. \tag{S20}$$

All partition functions are strictly positive, and the thermal purity is the purity of the Gibbs state $e^{-\beta H_L}/Z_L(\beta)$. The limiting spatial inequalities and the Boltzmann power sums give $0 < P_s, P_\tau \le 1$ at even lengths and

$$\begin{aligned} 1 - P_s(2n, \beta) &\le f(1 - P_s(n, \beta)), && \text{second line of Eq. (S16), divided by } 2Z_{2n}(\beta)^2, \\ 1 - P_\tau(L, 2\beta) &\le f(1 - P_\tau(L, \beta)), && \text{Eq. (S14) on the Boltzmann weights.} \end{aligned} \tag{S21}$$

Equation (S21) doubles n at fixed β or β at fixed L, but one recurrence step must double both; an exact identity between tori supplies the missing move. Gluing four copies into the doubled torus in either order gives the exact cancellation

$$P_s(n, 2\beta) P_\tau(n, \beta)^2 = \frac{Z_{2n}(2\beta)}{Z_n(\beta)^4} = \frac{\square\square}{\square^4} = P_s(n, \beta)^2 P_\tau(2n, \beta). \tag{S22}$$

Solving for one factor and repeating at length $2n$ (substitute Eq. (S20)) yields ✓L ✓L

$$P_{\mathrm{s}}(n,2\beta) = \frac{P_{\mathrm{s}}(n,\beta)^2 P_\tau(2n,\beta)}{P_\tau(n,\beta)^2},$$
$$P_\tau(4n,\beta) = \frac{P_\tau(2n,\beta)^2 P_{\mathrm{s}}(2n,2\beta)}{P_{\mathrm{s}}(2n,\beta)^2}. \tag{S23}$$

Dropping each denominator, a purity at most one, turns the identity into a lower bound on the purity the recurrence lacks.

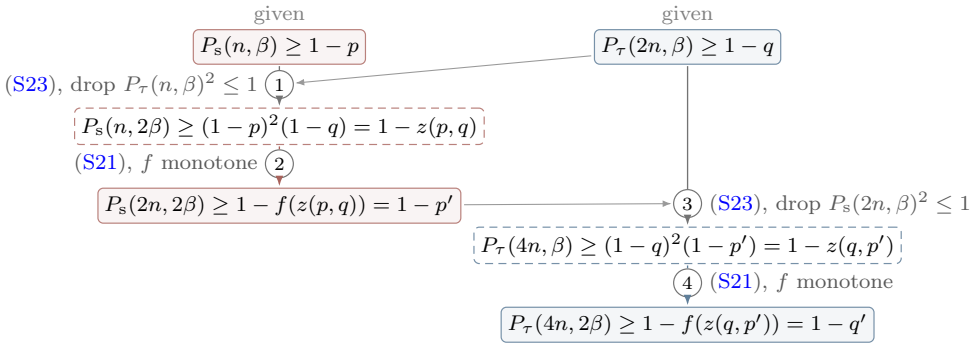
Let $n \geq 2$ be even, and let $0 \leq p, q < 1$ bound the seed purities:

$$P_{\mathrm{s}}(n,\beta) \geq 1-p, \qquad P_\tau(2n,\beta) \geq 1-q. \tag{S24}$$

To double both n and β (▫▭▭), write $z(a,b) = 1-(1-a)^2(1-b)$. The asymmetric update is ✓L

$$p' = f(z(p,q)), \qquad q' = f(z(q,p')). \tag{S25}$$

Assuming $p' < 1$, one doubling step advances (p,q) to (p',q') in four moves across space and imaginary time (thin arrows mark the second input at each torus step):



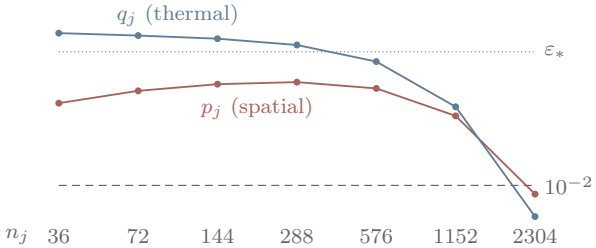
Move 3 substitutes the updated spatial bound p', not p, into the thermal step, so a small p' pulls q' down even when q starts above the symmetric fixed point ε_* of Sec. SI.E. Any upper bound below one may be substituted for p' in Move 3.

SI.E. Six rational steps from the seed to $[0, 1/100]^2$

The analytic tail estimate in Sec. SI.F requires both p and q to be at most $1/100$ (there one step costs at most $5\varepsilon^2$). The seed bounds of Sec. SII at $(n_0, \beta_0) = (36, 49/4)$,

$$P_{\mathrm{s}}(36, 49/4) \geq 1 - \frac{47916}{10^6}, \qquad P_\tau(72, 49/4) \geq 1 - \frac{181521}{10^6}, \tag{S26}$$

have larger (p_0, q_0). Moreover $q_0 = 0.1815$ exceeds the fixed point $\varepsilon_* \simeq 0.1273$ of the symmetric map $\varepsilon \mapsto f(1-(1-\varepsilon)^3)$, so a symmetric recurrence cannot start from this seed. We therefore iterate the asymmetric update (S25) six times, to $(n_6, \beta_6) = (36 \cdot 2^6, \frac{49}{4}2^6) = (2304, 784)$. Iterating f exactly squares the denominators at every step, so the exact orbit is not a usable certificate. Following Ref. [10], each update is rounded upward to seven decimal places: with $D = 10^7$, the stored integer $a_{j+1} = Dp_{j+1}$ is the ceiling of the exact value Df, $a_{j+1} - 1 \xrightarrow{Df} a_{j+1}$, and likewise for q. Rounding up keeps every bound valid, and the kernel checks the twelve rational inequalities, each exact update at most its stored entry. Table S1 lists the resulting sequence, drawn here on a logarithmic scale:



The spatial bound p_j first rises, to 0.0714 at $j = 3$, because its input $z(p, q) \geq q$ stays large while q does; once q drops below 0.15, both sequences fall rapidly. The thermal bound q_j falls from the first step, although it starts above ε_* (dotted): the updated p' in Move 3 keeps $f(z(q, p')) < q$ at every step.

TABLE S1. Six steps of the asymmetric recurrence (S25) along the dyadic scales $n_j = 36\,2^j$ and $\beta_j = (49/4)2^j$, starting from the seed bounds (S26) at $j = 0$. Column p_j bounds $1 - P_s(n_j, \beta_j)$ and column q_j bounds $1 - P_\tau(2n_j, \beta_j)$, each rounded upward to seven decimal places.

j	n_j	β_j	p_j	q_j
0	36	49/4	0.0479160	0.1815210
1	72	49/2	0.0605003	0.1733838
2	144	49	0.0686630	0.1632461
3	288	98	0.0713693	0.1447318
4	576	196	0.0633146	0.1055663
5	1152	392	0.0376144	0.0446518
6	2304	784	0.0084709	0.0055149

At $j = 6$, Table S1 gives ✓L

$$P_s(2304, 784) \geq 1 - 0.0084709 > 0.99,$$
$$P_\tau(4608, 784) \geq 1 - 0.0055149 > 0.99, \tag{S27}$$

with both p_6 and q_6 below 1/100, the entry condition of Sec. SI.F.

SI.F. Quadratic tail, spatial interpolation, and the uniform gap

Rounded rational steps cannot continue to arbitrarily large j, so the remaining steps need a closed-form bound. With $p, q \leq \varepsilon$ the input $z(p, q)$ is at most $1 - (1 - \varepsilon)^3$ and f is quadratic near zero, so one scalar recurrence bounds every further step. For $0 \leq \varepsilon \leq 1/100$ ✓L,

$$f(1 - (1 - \varepsilon)^3) \leq \frac{9\varepsilon^2}{2(1 - 3\varepsilon)^2} \leq 5\varepsilon^2 \leq \varepsilon. \tag{S28}$$

The first inequality uses $1 - (1 - \varepsilon)^3 \leq 3\varepsilon$ and the monotonicity of f, the second uses $1 - 3\varepsilon \geq 97/100$ so that $9/[2(97/100)^2] < 5$, and the third uses $5\varepsilon \leq 1/20 \leq 1$. With $p, q \leq \varepsilon$, the spatial update is therefore at most $5\varepsilon^2 \leq \varepsilon$; the thermal coupling input is then again at most $1 - (1 - \varepsilon)^3$, so the thermal update is at most $5\varepsilon^2$ as well.

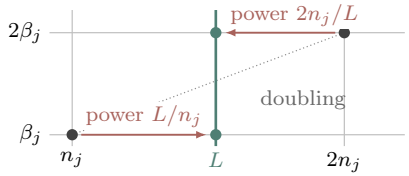
Starting from Eq. (S27), reset the scale index and put

$$n_j = 2304\,2^j, \qquad \beta_j = 784\,2^j, \qquad \varepsilon_j = \frac{20^{-2^j}}{5}, \qquad j \geq 0. \tag{S29}$$

The step $\varepsilon \mapsto 5\varepsilon^2$ squares 5ε, and $5\varepsilon_0 = 1/20$ at the threshold: hence the base 20, and $\log_{10} \varepsilon_j$ falls doubly exponentially from -2 at $j = 0$ to -21.5 at $j = 4$ ( ). Induction using Eq. (S28) proves ✓L

$$P_s(n_j, \beta_j) \geq 1 - \varepsilon_j, \qquad P_\tau(2n_j, \beta_j) \geq 1 - \varepsilon_j. \tag{S30}$$

Equation (S30) holds only at the dyadic scales, and an even L between them has no torus bound of its own. To cover every even length $L \geq 2304$, choose j with $n_j \leq L \leq 2n_j$. The two known tori supply the missing bound, from above at β_j and from below at $2\beta_j$:



The power-sum inequality in Eq. (S16) applies twice, with the powers v/u drawn above. At β_j take $(u, v) = (n_j, L)$; at $2\beta_j$ take $(u, v) = (L, 2n_j)$ and solve for the unknown torus:

$$Z_L(\beta_j)^2 \le Z_{n_j}(\beta_j)^{2L/n_j},$$
$$Z_L(2\beta_j) \ge Z_{2n_j}(2\beta_j)^{L/(2n_j)}. \tag{S31}$$

Set $r = L/(2n_j) \in [1/2, 1]$. Let x be the four-copy ratio of Eq. (S22) at scale j,

$$x = \frac{Z_{2n_j}(2\beta_j)}{Z_{n_j}(\beta_j)^4} = \frac{\boxplus}{\boxplus^4} = P_\tau(2n_j, \beta_j) P_s(n_j, \beta_j)^2.$$

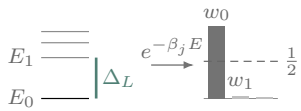
By Eq. (S30), $(1-\varepsilon_j)^3 \le x \le 1$, so dividing the second inequality in Eq. (S31) by the first yields

$$P_\tau(L, \beta_j) \ge x^r \ge x \ge (1-\varepsilon_j)^3 \ge 1 - 3\varepsilon_j > \frac{1}{2}. \tag{S32}$$

Every even $L \ge 2304$ thus has thermal purity above $1/2$ at a temperature β_j of its own scale.

Finally, we convert this thermal purity bound into a lower bound on the spectral gap. A purity above $1/2$ at one finite temperature bounds the largest Gibbs weight, and the ratio of the two largest weights is $e^{-\beta_j \Delta_L}$. Bounding that ratio strictly below 20^{-2^j} gives a strict gap independent of L. List the energies with multiplicity, $E_0(L) \le E_1(L) \le \cdots$. Each level m carries the Gibbs weight

$$w_m = \frac{e^{-\beta_j E_m(L)}}{Z_L(\beta_j)},$$



The purity is at most the largest weight, $\sum_m w_m^2 \le w_0$, so Eq. (S32) gives $w_0 > 1/2$. The ground state is therefore unique, with $w_1 \le 1 - w_0$. With Δ_L the gap drawn above, we obtain

$$e^{-\beta_j \Delta_L} = \frac{w_1}{w_0} \le \frac{3\varepsilon_j}{1-3\varepsilon_j}$$
$$\le \frac{60}{97}\, 20^{-2^j} < 20^{-2^j}, \tag{S33}$$

where $3\varepsilon_j = (3/5)20^{-2^j}$ and $1 - 3\varepsilon_j \ge 97/100$ give the prefactor $(3/5)(100/97) = 60/97$. Taking logarithms and dividing by $\beta_j = 784\, 2^j$ at this finite, length-dependent temperature gives ✓L

$$\boxed{\Delta_L > \frac{\log 20}{784} \quad \text{for every even } L \ge 2304}. \tag{S34}$$

The strict inequality comes from the finite prefactor $60/97 < 1$, without taking a fixed-length limit $\beta \to \infty$. On the jth dyadic interval, Eq. (S33) gives the larger bound $\frac{1}{\beta_j} \log \frac{1-3\varepsilon_j}{3\varepsilon_j}$, which decreases from 4.43×10^{-3} at $j = 0$ to 3.86×10^{-3} at $j = 4$ toward the uniform bound $\frac{\log 20}{784} \approx 3.82 \times 10^{-3}$ (, dashed).

SII: Seed bounds and gap estimates for the spin-1 chain

The unformalized computer-assisted proof of Ref. [10] establishes two periodic spin-1 gap bounds from two sets of finite inputs (in this section we write $b_0 = \beta_0$ and $b = r b_0$ for the seed inverse temperatures). First, short-ring trace enclosures on $4 \le n \le 12$ sites at $b_0 = 49/4$ together with a 72-site matrix-product trial state certify the seed bounds at $(n_0, b_0) = (36, 49/4)$ (Lemma 1) ✓L,

$$1 - P_s(36, 49/4) \le \frac{47916}{10^6},$$
$$1 - P_\tau(72, 49/4) \le \frac{181521}{10^6}, \tag{S35}$$

from which six asymmetric doubling steps reach $(2304, 784)$ and prove $\Delta_L > \frac{\log 20}{784} \simeq 0.003821$ for every even $L \geq 2304$ (Sec. SII.A). Second (Sec. 2.2 of Ref. [10]), a separate table of nine traces on $n \in \{6, 8, 10\}$ sites at $b_0' = 21/2$ combined with a 120-site trial state and a temperature lift $r = 5/2$ to $b = 105/4$ certifies $1 - P_{\mathrm{s}}(60, 105/4)$, $1 - P_{\tau}(120, 105/4) < 1/8$, proving $\Delta_L > \frac{4}{105} \log \frac{80}{79} \simeq 0.000479$ for every even $L \geq 60$.

We first independently reproduced and verified all of OpenAI's finite trace and matrix-product trial contractions in exact arithmetic. Section SII.A reviews the derivation of Eq. (S35) at $(36, 49/4)$. Section SII.B then adapts the temperature-lifting and threshold-refinement mechanism from Sec. 2.2 of Ref. [10] to the stronger $b_0 = 49/4$ ($4 \leq n \leq 12$) trace table together with the 120-site trial state at $r = 17/8$ ($b = 833/32$), tighter twelve-decimal polynomial caps U, V, and per-interval gap evaluation across ten doubling steps plus $[36, 72]$ from $(36, b_0)$. From the same verified finite inputs, this proves a unique ground state and ✓L

$$\Delta_L > 0.072901771 \quad \text{for every even } L \geq 36. \tag{S36}$$

Following Ref. [10], intermediate bounds are rounded outward to rationals so that every step reduces to exact rational comparisons. In Lean 4, we formalize the deduction from the short-ring trace table to Eq. (S35) and the doubling chains behind Eqs. (S35) and (S36); the lifted seed bounds of Table S3 and the twelve-decimal bridge seeds enter the Lean theorem for Eq. (S36) as hypotheses, verified in exact rational arithmetic outside Lean. We further verify the 72- and 120-site integer matrix-product trial contractions and the $n = 4$ trace totals directly in the Lean kernel, reducing the remaining hypotheses to the printed integer trace totals for $5 \leq n \leq 12$ ✓L.

SII.A. Seed purities at length 36 and inverse temperature 49/4

The seed inverse temperature $b_0 = 49/4$ is the one chosen in Ref. [10]. At this scale the two ratios

$$P_{\mathrm{s}}(36, b_0) = \frac{Z_{72}(b_0)}{Z_{36}(b_0)^2} = \frac{\square\square\square}{\square^2}, \qquad P_{\tau}(72, b_0) = \frac{Z_{72}(2b_0)}{Z_{72}(b_0)^2} = \frac{\square\square\square}{\square\square\square^2} \tag{S37}$$

need lower bounds on their numerators and upper bounds on their denominators. A 72-site trial state and exact traces on rings of 4 to 12 sites supply them:

	Partition function	Bound	Supplied by
▭	$Z_{72}(b_0)$	lower	trial state, through $\lvert\lambda_0\rvert > \ell$ (Sec. SII.A.d)
▭	$Z_{36}(b_0)$	upper	traces, through the bounds U, V (Sec. SII.A.c)
▭	$Z_{72}(2b_0)$	lower	trial state (Sec. SII.A.b)
▭	$Z_{72}(b_0)$	upper	traces, through the bounds U, V

a. *Shifted Hamiltonian, twisted rings, and the spatial transfer operator*

The seed ratios (S37) need partition functions at 36 and 72 sites, while certified traces stop at 12 sites. Three objects connect them: a Hamiltonian shift that turns the trial bound into $Z > 1$, twisted rings that label spin-rotation sectors, and a transfer operator whose eigenvalues govern every ring length. We consider the periodic spin-1 Heisenberg Hamiltonian

$$H_n = \sum_{j=1}^{n} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}, \qquad \boldsymbol{S}_j^2 = 2I, \qquad \boldsymbol{S}_{n+1} = \boldsymbol{S}_1, \tag{S38}$$

with unit exchange coupling. Lengths count sites, and $\Delta_L = E_1(L) - E_0(L)$, with eigenvalues ordered with multiplicity. We shift the Hamiltonian by κ per site, a rational with $-\kappa$ just above the trial energy per site of Sec. SII.A.b. The shifted ground energy is then negative and the leading transfer eigenvalue sits near one:

$$\kappa = \frac{700741}{500000} \simeq 1.401482, \qquad \widehat{H}_n = H_n + \kappa n I, \qquad Z_n(\beta) = \operatorname{Tr} e^{-\beta \widehat{H}_n}. \tag{S39}$$

We drop the hats of Eq. (9): in this section Z_n and K_β denote the shifted quantities. This extensive shift cancels in both purities ✓L:

$$P_{\mathrm{s}}(n, \beta) = \frac{Z_{2n}(\beta)}{Z_n(\beta)^2}, \qquad P_{\tau}(n, \beta) = \frac{Z_n(2\beta)}{Z_n(\beta)^2}. \tag{S40}$$

TABLE S2. Shifted short-ring traces $Z_n(b_0) = \operatorname{Tr} e^{-b_0(H_n+\kappa n I)}$ and twisted traces $Z_n(b_0, P) = \operatorname{Tr} e^{-b_0(H_n(P)+\kappa n I)}$ at $b_0 = 49/4$ and $\kappa = 700741/500000$ from Ref. [10], where P is a π spin rotation on the closing bond. Each tabulated rational center is certified within absolute error $\epsilon_Z = 3 \times 10^{-5}$ and supplies the sector moments J_n, N_n in Eq. (S50).

n	$Z_n(b_0)$ center	$Z_n(b_0, P)$ center	n	$Z_n(b_0)$ center	$Z_n(b_0, P)$ center
4	124.885379	0.065757	9	0.195455	1.514846
5	0.002925	2.960742	10	2.654823	0.675419
6	12.870614	0.286599	11	0.377629	1.314544
7	0.054069	1.920414	12	1.900547	0.787405
8	4.639283	0.510140			

To separate the spatial transfer spectrum by spin-rotation symmetry, we also use rings whose closing bond is twisted by a spin rotation, 

, alongside the plain ring  . For an on-site spin rotation u_g representing g, we define

$$H_n(g) = \sum_{j=1}^{n-1} h_{j,j+1} + \sum_{\alpha=x,y,z} S_n^\alpha (u_g S^\alpha u_g^\dagger)_1,$$
$$Z_n(\beta, g) = \operatorname{Tr} e^{-\beta(H_n(g)+\kappa n I)}, \tag{S41}$$

where $h_{j,j+1} = \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}$.

The continuous-time construction of Ref. [10] provides a self-adjoint one-site spatial transfer operator K_β for the shifted Hamiltonian and a virtual symmetry operator U_g for the twist. The twisted trace is then a ring of n transfer vertices closed through U_g:

$$Z_n(\beta, g) = \operatorname{Tr}(K_\beta^n U_g) = \text{  }, \qquad n \geq 4. \tag{S42}$$

At $g = 1$ the trace is the nth power sum of the signed spectrum  of K_β. Its two signs nearly cancel at odd n, so the odd traces in Table S2 are small ( on a logarithmic scale) and both parities are needed. Because the same eigenvalues govern every length $n \geq 4$, trace bounds on $4 \leq n \leq 12$ constrain the power sums at n_0 and $2n_0$.

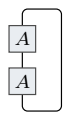
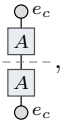
b. *Short-ring traces and the matrix-product trial state*

Two numerical inputs from Ref. [10] supply the short-ring moments and ground-state energy bound at $b_0 = 49/4$. The first input is the table of shifted traces $Z_n(b_0)$ and $Z_n(b_0, P)$ for $4 \leq n \leq 12$, where P is a π rotation about a Cartesian axis. Table S2 lists the rational centers, each certified within the absolute error

$$\epsilon_Z = 3 \times 10^{-5}, \tag{S43}$$

the error bound of the integer recursion described next.

The exact integer recursion of Ref. [10] writes each trace as $\operatorname{Tr} A^2$ with A real symmetric and folds it into squared columns,

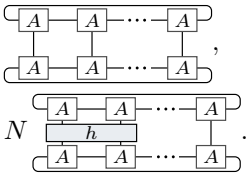
$$Z_n(b_0, g) = \text{  } = \sum_c \text{  }, \qquad A = e^{-b_0(H_n(g)+\kappa n I)/2}, \tag{S44}$$

so the recursion runs at $b_0/2$, one truncated Poisson polynomial of a rescaled Hamiltonian of norm at most one per column, in scaled integer arithmetic; the rounding errors add column by column over symmetry representatives with their multiplicities, within total error ϵ_Z.

The numerators need lower bounds at 72 sites, which no certified trace reaches; a variational state supplies them once its energy per site lies below $-\kappa$. The second input is a periodic matrix-product trial state ψ_N of bond dimension 26 with integer tensors A, used at $N = 72$ and at $N = 120$. Its norm and its energy, with h on one of the N bonds,

are exact integer contractions,

$$\|\psi_N\|^2 = \quad , \qquad \langle \psi_N, H_N \psi_N \rangle = N \quad . \tag{S45}$$



Exact integer arithmetic (verified in the Lean kernel for both $N = 72$ ✓L and $N = 120$ ✓L) verifies that for $N \in \{72, 120\}$,

$$\frac{\langle \psi_N, H_N \psi_N \rangle}{N \, \|\psi_N\|^2} < -\kappa, \tag{S46}$$

so the shifted ground energy $E_0(N) + \kappa N$ is negative by the variational principle. Retaining only the ground-state Boltzmann weight gives the lower bounds ✓L

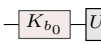
$$Z_{72}(\beta) > 1, \qquad Z_{120}(\beta) > 1 \quad (\beta > 0). \tag{S47}$$

In the base seed, $Z_{72}(2b_0) > 1$ bounds the thermal numerator directly, whereas $Z_{72}(b_0) > 1$ forces a large leading spatial eigenvalue.

c. *Spin-rotation sectors and polynomial eigenvalue bounds*

Bounding all eigenvalues of K_{b_0} from the untwisted traces alone would give the subleading eigenvalues the same bound $U > 1$ as the leading one, leaving the 36-site remainder too large. Instead, we split the spectrum of K_{b_0} into spin-rotation sectors so that the subleading sectors carry a strictly smaller bound $V < 1$. The four virtual π rotations $D_2 = \{1, P_x, P_y, P_z\}$ commute with K_{b_0},

$$\quad = \quad , \qquad g \in D_2, \tag{S48}$$



and define the character projectors

$$\Pi_\chi = \frac{1}{4} \sum_{g \in D_2} \chi(g) U_g, \qquad \mathrm{Tr}(K_{b_0}^n \Pi_\chi) = \quad . \tag{S49}$$



Cyclic permutation of the coordinate axes conjugates the three nontrivial characters into one another, so their three sectors share a common eigenvalue list N, while the D_2-invariant sector has eigenvalue list J. Writing $J_n = \sum_{\lambda \in J} \lambda^n$ and $N_n = \sum_{\lambda \in N} \lambda^n$ for their nth power sums at b_0, the character table of the plain and P_z-twisted rings,

	J	N_x	N_y	N_z	
	+	+	+	+	$Z_n = J_n + 3N_n$
π	+	−	−	+	$Z_n(P) = J_n - N_n$

identical spectra

inverts to the sector moments:

$$J_n = \frac{Z_n + 3Z_n(P)}{4}, \qquad N_n = \frac{Z_n - Z_n(P)}{4}. \tag{S50}$$

The inversion maps the trace error (S43) to an error at most ϵ_Z on every sector moment for $4 \le n \le 12$. Adding ϵ_Z to both twelfth-moment centers of Table S2 gives the upper bounds

$$J_{12} \le m_J = \frac{2131441}{2000000}, \qquad N_{12} \le m_N = \frac{556631}{2000000}. \tag{S51}$$

The twelfth moment alone bounds a J eigenvalue only by $m_J^{1/12} \simeq 1.0053$, whose 72nd power exceeds 1.46; the thermal denominator below needs a far tighter upper bound. A polynomial that is nonnegative on $\mathbb{R}$ makes its sector

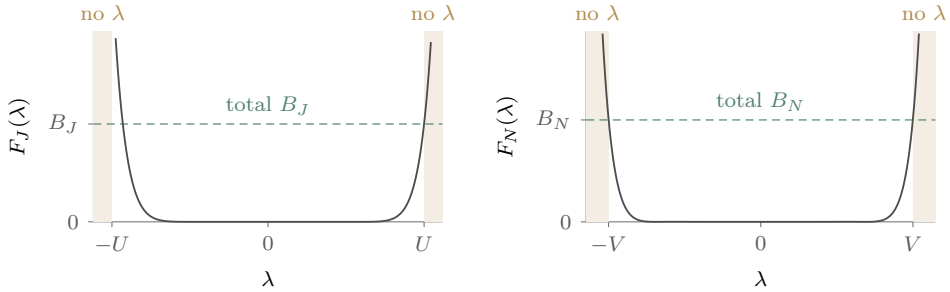
sum an upper bound for each eigenvalue. One that nearly vanishes on the subleading spectrum draws its total from the leading eigenvalue alone, tightening the bound. Ref. [10] takes degree at most 12, a fourth power times a square:

$$F_J(x) = x^4(7 + 17x - 280x^2 - 185x^3 + 1000x^4)^2,$$
$$F_N(x) = x^4(12 - 394x^2 + 1000x^4)^2. \tag{S52}$$

Expanded, each polynomial is a combination $\sum_j f_j x^j$ of $x^4, \ldots, x^{12}$ (F_N of the even powers only), so its sum over a sector is the same combination of the sector moments. Taking each moment at the end of its ϵ_Z interval that bounds its term (upper for $f_j > 0$, lower for $f_j < 0$) gives the certified totals

$$\sum_{\lambda \in J} F_J(\lambda) \le B_J = 318604.54848175, \qquad \sum_{\lambda \in N} F_N(\lambda) \le B_N = 48169.880699. \tag{S53}$$

Each polynomial rises above its total outside a bounded interval:



Each equation $F(x) = B$ has one positive root, $x_J \simeq 1.0013506440113$ for J and $x_N \simeq 0.8712557046526$ for N; the negative root for J has the smaller modulus, near 0.9315. For the base seed, Ref. [10] rounds the roots up to the short rationals $U = 1.00139$ and $V = 0.872$. Shifting $x = \pm(U + y)$ with $y \ge 0$ gives polynomials with positive coefficients, so Lean checks the bounds (with its own, slightly tighter rational caps) without root extraction, ruling out both exterior rays ✓L:

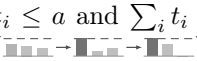
$$|\lambda| < U \quad (\lambda \in J), \qquad |\lambda| < V \quad (\lambda \in N). \tag{S54}$$

Section SII.A.d uses U in the thermal denominator and V for the three N sectors of both residuals.

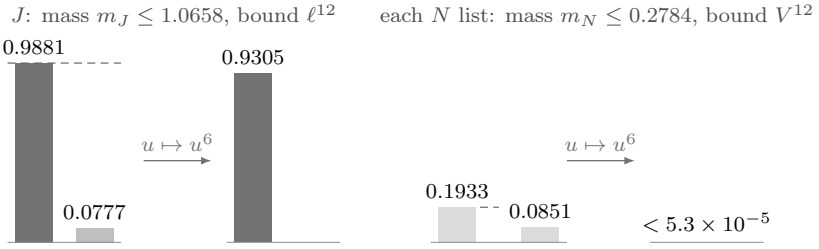
d. *Purity bounds at lengths* 36 *and* 72 *from twelfth moments*

The seed ratios need power sums at $n = 36$ and $n = 72$, far beyond the certified rings. The twelfth moments (S51) and the eigenvalue bounds (S54) still bound every higher moment through one extremal function. For $m \ge 0$, $a > 0$, and $r \ge 1$, set ✓L

$$\mathcal{C}(m, a, r) = qa^r + (m - qa)^r, \qquad q = \lfloor m/a \rfloor. \tag{S55}$$

Whenever $0 \le t_i \le a$ and $\sum_i t_i \le m$, convexity of t^r lets mass move from a smaller entry to a larger one without lowering $\sum_i t_i^r$ (). The maximum therefore fills q entries to the bound a and leaves $m - qa$ in one entry, so $\sum_i t_i^r \le \mathcal{C}(m, a, r)$. Applied to $t_i = \lambda_i^{12} \ge 0$, this bounds higher moments of the twelfth-power weights.

The spatial numerator is bounded by the leading eigenvalue alone, so that eigenvalue must also dominate the 36-site denominator; the trial bound $Z_{72}(b_0) > 1$ constrains only a sum. We first show that J contains a dominant eigenvalue with $|\lambda_0| > \ell = 0.999$, the short rational of Ref. [10]; any ℓ that keeps the sum (S56) under one would serve. If every $\lambda \in J$ satisfied $|\lambda| \le \ell$, Eq. (S55) with $r = 6$ would place the twelfth-power masses as follows,



and give

$$Z_{72}(b_0) \le \mathcal{C}(m_J, \ell^{12}, 6) + 3\mathcal{C}(m_N, V^{12}, 6) < 0.931 < 1, \tag{S56}$$

contradicting the 72-site trial bound $Z_{72}(b_0) > 1$ in Eq. (S47). Because $m_N < \ell^{12}$, no N eigenvalue reaches ℓ in modulus, so J holds an eigenvalue λ_0 with $|\lambda_0| > \ell$ ( ). This is where the sector split is needed: the plain traces alone could not exclude three copies of λ_0 in N.

Removing $\lambda_0^{12} > \ell^{12}$ leaves residual twelfth moment at most $m_J - \ell^{12}$ in J, below U^{12}, so it counts as a single entry. For even $k \ge 12$, the subleading moment $\sum_{i\neq 0} \lambda_i^k$ is therefore at most

$$R_k = (m_J - \ell^{12})^{k/12} + 3\mathcal{C}(m_N, V^{12}, k/12). \tag{S57}$$

At $k = 36$ and $k = 72$, keeping only the leading eigenvalue in the spatial numerator, the trial bound (S47) in the thermal numerator, and R_k in both denominators gives the two base purity bounds:

$$\begin{aligned} P_{\mathrm{s}}(36, b_0) &= \frac{Z_{72}(b_0)}{Z_{36}(b_0)^2} \ge \frac{\lambda_0^{72}}{(|\lambda_0|^{36} + R_{36})^2} \ge \left(1 + \frac{R_{36}}{\ell^{36}}\right)^{-2} = 1 - p, \\ P_{\tau}(72, b_0) &= \frac{Z_{72}(2b_0)}{Z_{72}(b_0)^2} \ge \frac{1}{(U^{72} + R_{72})^2} = 1 - q. \end{aligned} \tag{S58}$$

The sizes entering these bounds are

k	$(m_J - \ell^{12})^{k/12}$	$3\,\mathcal{C}(m_N, V^{12}, k/12)$	R_k	leading term	bound on $1 - P$
36	$0.0777^3 < 0.00047$	$3(0.1933^3 + 0.0851^3) < 0.02352$	< 0.024	$\ell^{36} > 0.9646$	$p < 1 - (0.9646/0.9886)^2 < 0.048$
72	$0.0777^6 < 3 \times 10^{-7}$	$3(0.1933^6 + 0.0851^6) < 1.6 \times 10^{-4}$	$< 1.6 \times 10^{-4}$	$U^{72} < 1.1052$	$q < 1 - 1.10536^{-2} < 0.1816$

Evaluating Eq. (S58) gives ✓

$$p \simeq 0.0479158 \le \frac{47916}{10^6}, \qquad q \simeq 0.1815202 \le \frac{181521}{10^6}, \tag{S59}$$

which completes the proof of Eq. (S35). Section SII.B.c reuses these formulas with sharper constants as the bridge on $36 \le L \le 72$.

SII.B. Seed purities at length 60 and the bound $\Delta_L > 0.07290$

The gap bound of the doubling argument is inversely proportional to the seed inverse temperature, so a seed at $b = rb_0$ with $r > 1$ gives a larger constant if $1 - P_{\mathrm{s}}$ and $1 - P_\tau$ stay small. Ref. [10], Sec. 2.2, introduced this temperature lifting on a separate trace table at $b_0' = 21/2$ (rings of 6, 8, and 10 sites, twists $1, P, C$), with $r = 5/2$ and the 120-site trial state. That seed at $n_0 = 60$ proves $\Delta_L > \frac{4}{105} \log \frac{80}{79} \simeq 0.000479$ for even $L \ge 60$. Here we apply the same lift to the two-sector trace table at $b_0 = 49/4$ (Table S2), raised to $b = rb_0$ by a convexity inequality and combined with the 120-site trial bound of Eq. (S47), twelve-decimal polynomial caps, and per-interval gap bounds over ten doubling steps. The length $n_0 = 60$ is half the trial ring, and the $(36, b_0)$ bound covers $36 \le L \le 72$, so every even length $L \ge 36$ is reached with $\Delta_L > 0.072901771$.

a. *Spatial purity at length 60 and inverse temperature $b = rb_0$*

At 120 sites and inverse temperature $b = rb_0$ ($r \simeq 2$), the eigenvalue bound U enters raised to the power $120r \ge 240$. At that exponent the five-decimal rounding $U = 1.00139$ of Eq. (S54) inflates U^{120r} by nearly 1% relative to the root x_J. Rounding both roots up to twelve decimal places instead gives

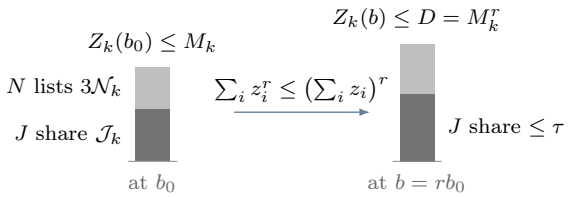
$$U = 1.001350644012, \qquad V = 0.871255704653. \tag{S60}$$

The step below needs trace bounds at b_0 on a ring of $k > 12$ sites (Table S3); the extremal moment bound supplies them. With m_J, m_N from Eq. (S51), since $U^{12} < m_J < 2U^{12}$ and $V^{12} < m_N < 2V^{12}$, Eq. (S55) bounds the sector moments at any even length $k \ge 12$ by

$$\mathcal{J}_k = U^k + (m_J - U^{12})^{k/12}, \qquad \mathcal{N}_k = V^k + (m_N - V^{12})^{k/12}, \qquad M_k = \mathcal{J}_k + 3\mathcal{N}_k. \tag{S61}$$

Because $N_k \geq 0$ at even k, these bound both traces, $Z_k(b_0) \leq M_k$ and $Z_k(b_0, P) \leq \mathcal{J}_k$.

Trace bounds exist only at b_0, so the step to $b = rb_0$ must act on the Boltzmann weights themselves. Fix a temperature ratio $r \geq 1$ and an even length $12 \leq k \leq n_0$ (here $n_0 = 60$ and $2n_0 = 120$). Both H_k and $H_k(P)$ are self-adjoint, so their Boltzmann weights $z_i > 0$ satisfy $\sum_i z_i^r \leq (\sum_i z_i)^r$. This raises the trace bounds from b_0 to b, and with them the J share of Eq. (S50):



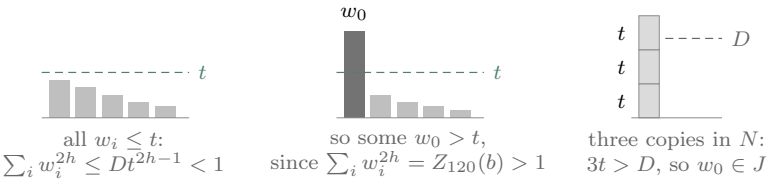
$$Z_k(b) \leq D := M_k^r, \qquad Z_k(b, P) \leq E := \mathcal{J}_k^r. \tag{S62}$$

The totals D and E are all the lifted argument knows at b; as at b_0, the trial bound must force one weight that dominates the n_0-site denominator. At b, let $w_i = \lambda_i(b)^k \geq 0$ be the kth powers of the eigenvalues of the one-site transfer operator (k is even). With $h = n_0/k$, so that w_i^h and w_i^{2h} are the n_0- and $2n_0$-site weights, Eq. (S62) and the trial bound $Z_{2n_0}(b) > 1$ of Eq. (S47) give

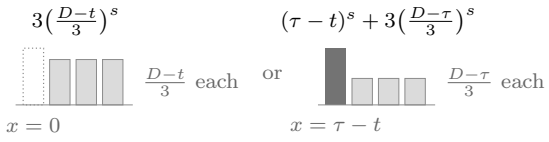
$$\sum_i w_i = Z_k(b) \leq D, \qquad \sum_{\lambda_i \in J} w_i = \frac{Z_k(b) + 3Z_k(b, P)}{4} \leq \tau := \frac{D + 3E}{4}, \tag{S63}$$

$$\sum_i w_i^{2h} = Z_{2n_0}(b) > 1. \tag{S64}$$

If a threshold $t > 0$ satisfies $Dt^{2h-1} < 1$ and $3t > D$, then some weight w_0 exceeds t and lies in J:



The n_0-site denominator is w_0^h plus the hth moment of the other weights, whose split between J and the N lists is unknown. Removing w_0 leaves residual J mass $x \in [0, \tau - t]$ and at most $(D - t - x)/3$ in each of the three identical N sectors. For $s \geq 1$ the residual moment $\sum_{i \neq 0} w_i^s \leq x^s + 3[(D - t - x)/3]^s$ is convex in x, so its maximum sits at an endpoint,



$$\mathcal{R}_s(t) = \max\left\{3\left(\frac{D-t}{3}\right)^s, (\tau - t)^s + 3\left(\frac{D-\tau}{3}\right)^s\right\}. \tag{S65}$$

All choices below satisfy $0 < t \leq \tau \leq D$.

Once $w_0 > t$ holds, any larger rational t' with $(t')^{2h} + \mathcal{R}_{2h}(t) < 1$ also satisfies $w_0 > t'$, since otherwise the $2n_0$-site sum would fall below one. Testing rationals avoids a $2h$th root, and iterating raises the threshold ( ). At $n_0 = 60$, the leading n_0-site weight w_0^h exceeds t^h and the rest sum to at most $\mathcal{R}_h(t)$, so the spatial purity satisfies

$$P_s(n_0, b) \geq \left(1 + \frac{\mathcal{R}_h(t)}{t^h}\right)^{-2}. \tag{S66}$$

Its complement is the p_0 column of Table S3.

TABLE S3. Certified seed bounds $p_0 \geq 1 - P_s(60, rb_0)$ and $q_0 \geq 1 - P_\tau(120, rb_0)$and resulting uniform gap bounds at length $n_0 = 60$ and inverse temperature $b = rb_0$ ($b_0 = 49/4$). Here k is the even ring length at which the trace bounds (S61) are raised to the power r in Eq. (S62). Running $K = 10$ doubling steps from $(60, rb_0)$ certifies the stated gap bound for all even $L \geq 60$ ✓L, and combining with the $(36, b_0)$ bound on $36 \leq L \leq 72$ extends it to all even $L \geq 36$ ✓L.

k	r	$p_0 \geq 1 - P_s$	$q_0 \geq 1 - P_\tau$	$\Delta_L >$
30	2	0.0307608679	0.0590740709	0.070450002
28	17/8	0.0305302052	0.0479556776	0.072901771

b. *Thermal purity at length* 120 *and inverse temperature* $b = rb_0$

Neither 120-site partition function in the thermal purity at b is certified. Both the trial bound and the moment bound (S61) concern the Boltzmann weights z_i of the shifted 120-site Hamiltonian at b_0; the lift to b raises each to the rth power. We therefore bound $P_\tau(2n_0, b)$ from the ordered weights at b_0 (): the trial state gives $z_0 > 1$ and Eq. (S61) gives $\sum_i z_i \leq M_{2n_0}$, hence

$$\sum_{i>0} \frac{z_i}{z_0} \leq M_{2n_0} - 1, \qquad \sum_{i>0} \left(\frac{z_i}{z_0}\right)^r \leq (M_{2n_0} - 1)^r. \tag{S67}$$

At $b = rb_0$ the weights become z_i^r, which concentrates them on the ground state ( at $r = 17/8$). Keeping only the ground term z_0^{2r} in the numerator $Z_{2n_0}(2b)$ and dividing by it gives the thermal purity bound at $2n_0 = 120$

$$P_\tau(2n_0, rb_0) \geq \left[1 + (M_{2n_0} - 1)^r\right]^{-2}. \tag{S68}$$

Its complement is the q_0 column of Table S3; with p_0 it seeds the recurrence of Sec. SII.B.c.

c. *Combining the dyadic intervals for all even* $L \geq 36$

The quadratic tail of Sec. SI.F needs both p and q below $1/100$, reached only after K doublings, so it covers only $L \geq n_K$; the K earlier doubling intervals need their own bounds. Write (p_j, q_j) for the bounds on $1 - P_s(n_j, b_j)$ and $1 - P_\tau(2n_j, b_j)$ after j of K steps of the asymmetric recurrence (S25) from the seed, with the last step below the threshold $1/100$ of Sec. SI.F:

$$(n_j, b_j) = (2^j n_0, 2^j b), \qquad A_j = (1 - p_j)^2(1 - q_j), \qquad u = \max(p_K, q_K) \leq \frac{1}{100}.$$

On the interval between two consecutive scales, the spatial interpolation inequality (S32) bounds the largest normalized Boltzmann weight a_0 by A_j, which exceeds $1/2$ at every step below. The ground state is therefore unique, the next weight is $a_1 \leq 1 - a_0$, and

$$a_0 \geq P_\tau(L, b_j) \geq A_j^{L/(2n_j)} \geq A_j, \qquad e^{-b_j \Delta_L} = \frac{a_1}{a_0} \leq \frac{1 - A_j}{A_j},$$
$$\Delta_L \geq \frac{\log(A_j/(1 - A_j))}{b_j} \quad (n_j \leq L \leq 2n_j,\ j < K). \tag{S69}$$

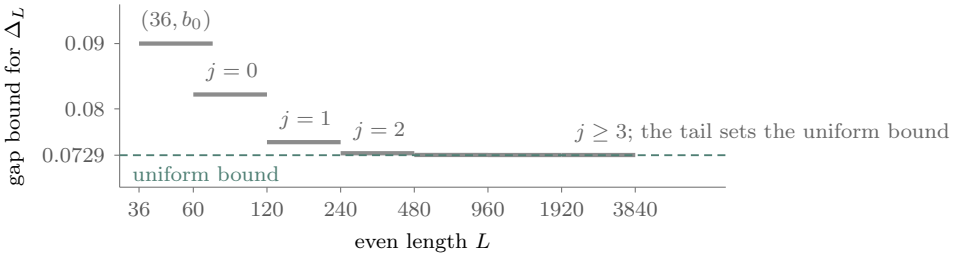
For even $L \geq n_K$, the quadratic tail of Sec. SI.F takes over with its exact coefficient $G(u) = f(1 - (1 - u)^3)/u^2$, which rises from $G(0) = 9/2$ to 4.69 at $u = 1/100$ (Sec. SI.F rounds it to 5). With a rational $C \geq G(u)$, the bounds (p_m, q_m) at scale $m \geq K$ are at most $C^{-1}(Cu)^{2^{m-K}}$, and $\Delta_L > \frac{\log(1/(Cu))}{b_K}$ (using $3u/(1 - 3u) \leq Cu$, which holds because $C \geq 9/2$ and $u \leq 1/100$). The minimum of the K interval bounds (S69) and this tail bound therefore bounds Δ_L for every even $L \geq n_0$.

For the seeds at $(60, rb_0)$, we run $K = 10$ doubling steps, enough for the per-interval bounds to converge.

Choosing r balances two effects. A larger r shrinks the thermal residual $(M_{120} - 1)^r$ in Eq. (S68) and increases $b = rb_0$. Too large an r, however, makes $D = M_k^r$ too large for a threshold t with $Dt^{2h-1} < 1$ and $3t > D$ to isolate a leading spatial eigenvalue.

Table S3 lists two choices near the optimum $r \approx 2$. The integer choice $(k, r) = (30, 2)$ has $h = 60/k = 2$, so Eq. (S65) uses only second and fourth powers; $(k, r) = (28, 17/8)$ maximizes the gap bound.

To cover the remaining even lengths $36 \le L < 60$, we evaluate the $(36, b_0)$ formulas (S57)–(S58) with the twelve-decimal bounds (S60) and $\ell = 0.999997937629$. This ℓ is the largest twelve-decimal value satisfying Eq. (S56), and its A_0 in Eq. (S69) gives $\Delta_L > 0.09$ on $36 \le L \le 72$. The intervals and their bounds for $(k, r) = (28, 17/8)$ are



(per-interval values are floating-point illustrations; the certified minimum is the dashed line). Because $[36, 72]$ and the dyadic intervals $[60\,2^j, 120\,2^j]$ overlap on $[60, 72]$, together they prove Eq. (S36) for every even $L \ge 36$.

SIII: Reflection-positive spin chains and ladders

Applying the periodic doubling argument of Section SI to a new Hamiltonian requires two inputs:

1. write $H_L = \sum_j h_{j,j+1} + \sum_j o_j$ in the real bond-reflection-positive form $-h = \sum_\alpha C_\alpha \otimes FC_\alpha F$ with $FoF = o$ on a chosen unit cell, so that the two-cell Trotter transfer matrix $T_{\beta,M} = \mathcal{F}^{\mathsf{T}}\mathcal{F} \succeq 0$ of Eq. (S9) is positive semidefinite at every even cell count L;
2. bound $1 - P_{\mathrm{s}}(n_0, \beta_0)$ and $1 - P_\tau(2n_0, \beta_0)$ at one finite scale by rationals (p_0, q_0) inside the convergence region of the doubling recurrence.

Reflection positivity carries no gap information (gapless and spontaneously dimerized chains are reflection positive too; Proposition S3), so the seed computation must be matched to the correlation length ξ and gap Δ of the model:

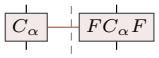
Regime	Finite inputs	Seed strategy	Models in Table I
Short ξ, small d^{2n_0}	$2n_0 \le 16$ spins	Symmetry blocks + Weyl bound	Staggered, dimerized, ladders, Ising (Sec. SIII.B)
Moderate ξ, large d^{4n_0}	6, 12 spins	Block spectra + Eq. (S22)	Single-ion anisotropy, $D = 4$ (Sec. SIII.B)
Weak off-diagonal term	6, 12 spins	Trial residuals + diagonal count	Mixed-field Ising, kernel seed (Sec. SIII.C)
Long ξ ($n_0 \ge 36$)	4–12 traces + MPS	Twisted-ring moments + integer MPS	Heisenberg, $S = 1$ (Sec. SII)

SIII.A. Reflection-positive spin chains, unit cells, and obstructions

To apply the finite-Trotter Gram factorization of Section SI.A to a periodic ring of $L \ge 2$ unit cells,

$$H_L = \sum_{j=1}^{L} h_{j,j+1} + \sum_{j=1}^{L} o_j, \qquad j + L \equiv j, \tag{S70}$$

we require a real local cell basis, a real symmetric involution F ($F^2 = I$), a real symmetric on-site matrix o satisfying $FoF = o$, and real symmetric or antisymmetric bond matrices C_α such that

$$-h = \sum_\alpha C_\alpha \otimes FC_\alpha F = \sum_\alpha$$ 

. (S71)

Hermiticity of h alone does not supply this form: the factors must be real and o even under F, so basis and cell are chosen model by model.

a. Bilinear, staggered, and biquadratic spin chains

For a bilinear bond, a factor $v \cdot \boldsymbol{S}$ and its mirror image must multiply back to $-\boldsymbol{S}^{\mathsf{T}} J \boldsymbol{S}$, so the mirror has to act on the spin components by a 3×3 matrix O. Only maps realized by an antiunitary symmetry qualify, which restricts the sign pattern of J. Consider first a single-site spin-S chain ($F = I$) with a real symmetric exchange matrix J and a uniform field $\boldsymbol{h}$,

$$H_L = \sum_{j=1}^{L} \boldsymbol{S}_j^{\mathsf{T}} J \boldsymbol{S}_{j+1} + \sum_{j=1}^{L} \boldsymbol{h} \cdot \boldsymbol{S}_j. \tag{S72}$$

In the standard S^z eigenbasis, S^x and S^z are real and S^y is purely imaginary, so complex conjugation K acts on the spin vector by $\mathrm{diag}(1, -1, 1)$. Composing K with a π rotation U_R about a principal axis gives an antiunitary reflection $\Theta = U_R K$ that acts on the spin vector by a symmetric orthogonal matrix $O = R\,\mathrm{diag}(1, -1, 1)$ with $\det O = -1$; the rotation about y gives time reversal $T = e^{i\pi S^y} K$ with $O = -I$.

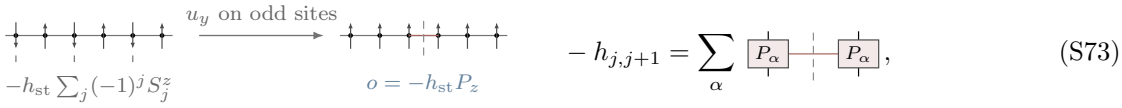
Proposition S1 (Bilinear spin chains)**.** *If there exists a symmetric orthogonal matrix $O \in \mathrm{O}(3)$ with $\det O = -1$ such that $-JO \succeq 0$ and $O\boldsymbol{h} = \boldsymbol{h}$, then Eq. (S72) admits a reflection-positive bond factorization for every spin S. For diagonal exchange $J = \mathrm{diag}(J_x, J_y, J_z)$ with nonzero entries and $\boldsymbol{h} = 0$, such an O exists if and only if an even number of J_x, J_y, J_z are negative.*

Proof. Factor the positive semidefinite matrix $-JO = \sum_\alpha v_\alpha v_\alpha^{\mathsf{T}}$ with real vectors $v_\alpha \in \mathbb{R}^3$ and set $C_\alpha = v_\alpha \cdot \boldsymbol{S}$. Under $\Theta = U_R T$, $\Theta(C_\alpha) = (O v_\alpha) \cdot \boldsymbol{S}$, so $\sum_\alpha C_\alpha \otimes \Theta(C_\alpha) = \boldsymbol{S}^{\mathsf{T}}(-JO)O\boldsymbol{S} = -\boldsymbol{S}^{\mathsf{T}} J \boldsymbol{S}$, while $\Theta(\boldsymbol{h} \cdot \boldsymbol{S}) = (O\boldsymbol{h}) \cdot \boldsymbol{S} = \boldsymbol{h} \cdot \boldsymbol{S}$. When J is diagonal with nonzero entries, $-JO \succeq 0$ forces $O = \mathrm{sign}(-J)$, whose determinant $(-1)^{n_-}$ is -1 if and only if the number n_- of ferromagnetic ($J_\alpha < 0$) components is even. □

Proposition S1 sorts the standard single-site spin chains by the sign parity n_- of their exchange couplings; a uniform field fails because $O = -I$ reverses it:

Chain	(J_x, J_y, J_z), $\boldsymbol{h}$	Mirror O	Reflection positive
Heisenberg antiferromagnet, any S	$(1, 1, 1)$, 0	$-I$ ($n_- = 0$)	yes
XXZ, $\Delta \geq 0$	$(1, 1, \Delta)$, 0	$-I$	yes
XY, either sign	$(J_x, J_y, 0)$, 0	$J_z = 0$ frees O_{zz}	yes
Transverse- or mixed-field Ising	$(0, 0, -1)$, $(g, 0, h)$	$\mathrm{diag}(1, -1, 1)$ fixes $\boldsymbol{h}$	yes
Heisenberg ferromagnet	$(-1, -1, -1)$, 0	$n_- = 3$	no
XXZ, $\Delta < 0$	$(1, 1, \Delta)$, 0	$n_- = 1$	no
Heisenberg antiferromagnet in a uniform field	$(1, 1, 1)$, $\boldsymbol{h} \neq 0$	$-I$ reverses $\boldsymbol{h}$	no

Although a uniform field breaks reflection positivity of the Heisenberg antiferromagnet, a *staggered* field restores it on every even ring: a π rotation about the y axis on every odd site makes the field uniform,



(S73)

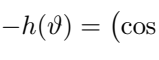
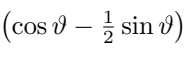
and the bond sign parity stays even, $n_- = 2$, because the remaining sign sits in the real antisymmetric factor $P_y = iS^y$. The same rotation underlies the staggered-field chain in Table I and, on open chains, relates opposite endpoint fields ↑•••••↓ to equal endpoint fields ↑•••••↑ (Section SV).

For $S = 1$, adding a biquadratic interaction gives the family

$$H_L(\vartheta) = \sum_{j=1}^{L} \Big[\cos\vartheta\, \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1} + \sin\vartheta\, (\boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1})^2 \Big]. \tag{S74}$$

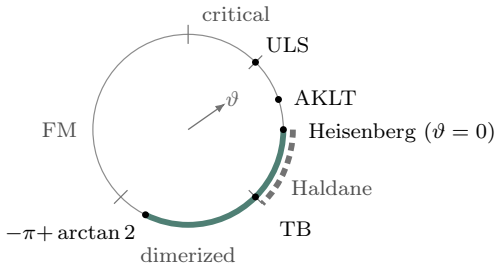
Proposition S1 does not reach the biquadratic term, which mixes products $S^\alpha S^\beta$ across the bond; factoring it needs one-site real operators without cross terms across the mirror. In the Cartesian basis $|x\rangle, |y\rangle, |z\rangle$ of Eq. (S2), the spin generators $G_\alpha = iS^\alpha$ are real antisymmetric and the products $Q_{\alpha\beta} = S^\alpha S^\beta$ are real matrices whose symmetric part

$Q^+_{\alpha\beta} = \frac{1}{2}(Q_{\alpha\beta} + Q_{\beta\alpha})$ and antisymmetric part $\frac{1}{2}\epsilon_{\alpha\beta\gamma}G_\gamma$ are orthogonal under the trace. Expanding $(\boldsymbol{S}_1 \cdot \boldsymbol{S}_2)^2$ into G_α and $Q^+_{\alpha\beta}$ diagonalizes the bond across the mirror:

$$-h(\vartheta) = \left(\cos\vartheta - \tfrac{1}{2}\sin\vartheta\right)\sum_\alpha$$  $$- \sin\vartheta \sum_{\alpha,\beta}$$  . (S75)

Proposition S2 (Reflection positivity of the bilinear–biquadratic chain)**.** *In the Cartesian basis, the bond* (S75) *is reflection positive if and only if both coefficients are nonnegative, that is,*

$$-\pi + \arctan 2 \le \vartheta \le 0. \tag{S76}$$



Reflection positivity of the whole family thus reduces to the signs of two coefficients; the frustration-free AKLT point ($\tan\vartheta = 1/3 > 0$) lies outside the arc because its positive biquadratic term has the wrong sign in Eq. (S75). The doubling argument applies only on the Haldane sub-arc $(-\pi/4, 0]$: the spontaneously dimerized phase $\vartheta \in [-\pi + \arctan 2, -\pi/4]$ and the gapless spin-1/2 Heisenberg chain are reflection positive but have no seed in the convergence region, for reasons no larger ring or lower temperature removes:

Proposition S3 (Obstructions to a convergent seed)**.** *For any reflection-positive ring Hamiltonian* (S70)*, no scale* (n_0, β_0) *has purities* $P_{\mathrm{s}}(n_0, \beta_0)$ *and* $P_\tau(2n_0, \beta_0)$ *inside the convergence region of the doubling recurrence if either*

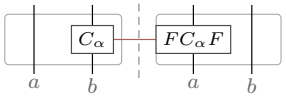
(i) H_{2n_0} *has a* k*-fold degenerate ground level with* $k \ge 2$*: the ordered Gibbs weights satisfy* $w_0 = w_1 \le 1/k \le 1/2$ *(*▮▮*), so* $P_\tau(2n_0, \beta_0) = \sum_m w_m^2 \le w_0 \le 1/2$*; or*

(ii) $\liminf_{L\to\infty,\ L\ even} \Delta_L = 0$*: a seed in the convergence region would give a uniform positive lower bound on* Δ_L *for all even* $L \ge n_0$*.*

Reflection positivity thus supplies the squaring map, and only a finite seed inside the convergence region separates gapped chains with a unique ground state from critical or symmetry-broken ones.

b. *Multi-spin unit cells, dimer reflections, and explicit decompositions*

When interactions alternate from bond to bond or couple two parallel legs, we block sites into a multi-spin unit cell. A single-site mirror then fails, because the inter-cell bond couples the last spin of one cell to the first spin of the next. The cell reflection F repairs this: across an inter-cell mirror cut, a reflection of the ring reverses the left-to-right order of the sites inside each cell:

$$-h_{j,j+1} = \sum_\alpha$$  . (S77)

In a two-spin dimer cell (a, b), F is the two-spin swap $F(x \otimes y) = y \otimes x$ ✓ʟ, which maps $C_\alpha = I \otimes G_\alpha$ on spin b to $FC_\alpha F = G_\alpha \otimes I$ on spin a of the next cell as drawn, and the intra-cell bond is a swap-invariant on-site matrix $FoF = o$. In a two-leg ladder the cell is a rung and the mirror keeps each leg on its own side,

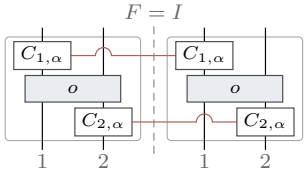
$$-h_{j,j+1} = \sum_\alpha$$  , (S78)

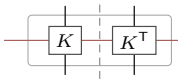
TABLE S4. Explicit reflection-positive cell decompositions $-h = \sum_\alpha C_\alpha \otimes FC_\alpha F$ and $FoF = o$ for the seven additional models in Table I, with links to their Lean 4 reflection-positivity proofs. Here F is the cell reflection involution, $u_y = e^{i\pi S^y}$ is a π spin rotation about y, $P_\alpha \in \{S^x, iS^y, S^z\}$ are the real spin-1/2 operators of Eq. (S80), $G_\alpha = iS^\alpha$ are the real antisymmetric spin-1 generators in the Cartesian basis, and $S_{ab}(x \otimes y) = y \otimes x$ swaps the two spins of a dimer cell.

Model	Unit cell (F)	Local basis / rotation	Bond operators C_α	On-site cell matrix o
Staggered field, $S = \frac{1}{2}$ ✓L	1 spin (I)	u_y on odd sites	P_α ($\alpha = x, y, z$)	$-h_{\mathrm{st}} P_z$
Dimerized, $S = \frac{1}{2}$ ✓L	2-spin dimer (S_{ab})	u_y on spin b	$\sqrt{1-\delta}\,(I \otimes P_\alpha)$	$-(1+\delta)\sum_\alpha P_\alpha \otimes P_\alpha$
Dimerized, $S = 1$ ✓L	2-spin dimer (S_{ab})	Cartesian basis	$\sqrt{1+\delta}\,(I \otimes G_\alpha)$	$-(1-\delta)\sum_\alpha G_\alpha \otimes G_\alpha$
Single-ion anisotropy, $S = 1$ ✓L	1 spin (I)	Cartesian basis	G_α ($\alpha = x, y, z$)	$D\,\mathrm{diag}(1, 1, 0)$
Heisenberg ladder, $S = \frac{1}{2}$ ✓L	2-spin rung (I)	checkerboard u_y	$P_\alpha \otimes I,\ I \otimes P_\alpha$	$-J_\perp \sum_\alpha P_\alpha \otimes P_\alpha$
Ising ladder ✓L	2-spin rung (I)	σ^z basis	$\sigma^z \otimes I,\ I \otimes \sigma^z$	$-\sigma^z \otimes \sigma^z - g(\sigma^x \otimes I + I \otimes \sigma^x)$
Mixed-field Ising ✓L	1 spin (I)	σ^z basis	σ^z	$-g\sigma^x - h\sigma^z$

so the rung interaction is even under the reflection and belongs to o. These two reflections, the dimer swap and the identity on a rung, are all that Table S4 needs.

The two-cell Gram form makes $Z_n(\beta)$ a sum of nonnegative powers only at even n, so an odd seed length needs positivity of the one-cell transfer operator itself. A dimer cell whose intra-cell bond is reflection positive supplies it: cutting the cell across its internal bond factors the *one-cell* spatial transfer operator into a half-cell product with its transpose,

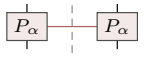
$$X_\beta = \quad = KK^{\mathsf{T}} \succeq 0. \tag{S79}$$



Every eigenvalue of KK^{T} is nonnegative, so $Z_n(\beta) = \mathrm{Tr}\, X_\beta^n = \sum_i \lambda_i(\beta)^n$ is a sum of nonnegative powers at *odd* cell counts n_0 too. In the dimerized spin-1 chain at $\delta = 7/10$, Eq. (S79) allows an odd seed of $n_0 = 3$ dimer cells (6 and 12 spins), avoiding the 16-spin ($3^{16} \simeq 4.3 \times 10^7$) ring that an even seed $n_0 = 4$ would require.

Proposition S1 guarantees a factorization, but the exact checkers behind Table S4 need its factors as explicit real matrices, and S^y is imaginary in the S^z basis. For $S = 1/2$, we write $P_x = S^x = \frac{1}{2}\sigma^x$, $P_y = iS^y = \frac{1}{2}(i\sigma^y)$, and $P_z = S^z = \frac{1}{2}\sigma^z$, which are real matrices satisfying $P_x^{\mathsf{T}} = P_x$, $P_y^{\mathsf{T}} = -P_y$, and $P_z^{\mathsf{T}} = P_z$. Rotating one site of an antiferromagnetic bond by $u_y = e^{i\pi S^y} = 2P_y$ maps $(S^x, S^y, S^z) \mapsto (-S^x, S^y, -S^z)$ and converts $-\boldsymbol{S}_1 \cdot \boldsymbol{S}_2$ into the real reflection-positive form ✓L

$$-u_{y,2}(\boldsymbol{S}_1 \cdot \boldsymbol{S}_2)u_{y,2}^\dagger = \sum_{\alpha=x,y,z} P_\alpha \otimes P_\alpha = \sum_{\alpha=x,y,z} \quad . \tag{S80}$$



Applying u_y on every odd site of an even spin-1/2 chain (or on the second spin b of every dimer cell, or on the checkerboard sublattice of a two-leg ladder for even L) is a unitary transformation that preserves the spectrum ✓L and puts every antiferromagnetic bond into the form (S80). For $S = 1$, we work directly in the Cartesian basis $|x\rangle, |y\rangle, |z\rangle$ of Eq. (S2), where $G_\alpha = iS^\alpha$ is real antisymmetric, $-\boldsymbol{S}_1 \cdot \boldsymbol{S}_2 = \sum_\alpha G_\alpha \otimes G_\alpha$, and $(S^z)^2 = \mathrm{diag}(1, 1, 0)$ is real diagonal without any sublattice rotation. Table S4 lists the resulting cell, reflection F, bond operators C_α, and on-site matrix o for all seven additional models in Table I.

SIII.B. Seed bounds from verified finite-ring block spectra

Once a model is in the reflection-positive form (S71), the doubling recurrence of Section SI applies. For the first six models in Table I, the correlation length is short enough that the seed scale n_0 lies between 2 and 12 unit cells. Even there, the upper bounds (p_0, q_0) need two-sided enclosures of three partition functions, and a dense eigensolve returns only floating-point numbers whose accuracy nothing certifies. The remedy is a posteriori: take the floating-point eigenbasis as a candidate and certify it by residual norms whose own rounding error is bounded.

To enclose the complete spectrum of H_L at $L \in \{n_0, 2n_0\}$ (up to 16 spin-1/2 or 12 spin-1 sites), we block-diagonalize H_L by translation, lattice reflection, and conserved spin or parity quantum numbers into self-adjoint blocks B_q whose exact integer multiplicities m_q sum to d^L, with every entry enclosed in rational or interval arithmetic. From a floating-point eigenbasis V and candidate eigenvalues $\tilde{\lambda}_i$ of B_q, we bound the Frobenius norms of the residual $R = B_q V - V\,\mathrm{diag}(\tilde{\lambda})$ and orthogonality error $G = V^{\mathsf{T}}V - I$ (including floating-point rounding error [42]); Ostrowski's

TABLE S5. Seed scales (n_0, β_0), certified starting bounds $p_0 \geq 1 - P_s(n_0, \beta_0)$ and $q_0 \geq 1 - P_\tau(2n_0, \beta_0)$, and resulting uniform gap bounds for all eight models in Table I (n_0 counts sites, two-spin dimer cells, or two-spin rungs as in Table I). The pairs (p_0, q_0) are four-digit upper bounds on the enclosures of the model certificates and enter the Lean theorems as hypotheses; the last row is the lifted seed $(k, r) = (28, 17/8)$ of Table S3. The printed gap is the interval bound of the certificate; each linked Lean theorem proves the rational-chain bound (S83) from its seed pair, which agrees in the quoted digits except for the single-ion row, whose Lean theorem starts from the direct seed $(6, 9/4)$ and proves 0.6946. The mixed-field Ising seed is proved unconditionally in the Lean kernel.

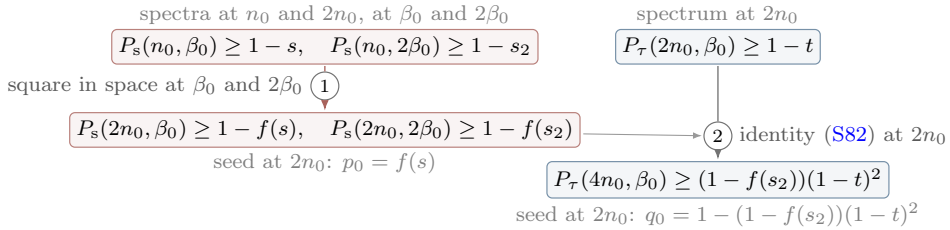
Model	Parameter	Seed (n_0, β_0)	$p_0 \geq 1 - P_s(n_0, \beta_0)$	$q_0 \geq 1 - P_\tau(2n_0, \beta_0)$	Certified gap $\geq$
Staggered field, $S = \frac{1}{2}$	$h_{st} = 1$	$(8, 21/4)$	6.937×10^{-4}	7.327×10^{-4}	1.1489 ✓L
Dimerized, $S = \frac{1}{2}$	$\delta = 1/2$	$(4, 6)$	1.133×10^{-2}	1.224×10^{-2}	0.5347 ✓L
Dimerized, $S = 1$	$\delta = 9/10$	$(2, 7/2)$	4.972×10^{-2}	3.206×10^{-2}	0.5288 ✓L
Single-ion anisotropy, $S = 1$	$D = 4$	$(12, 29/10)$	3.633×10^{-3}	3.141×10^{-2}	0.8994 ✓L
Heisenberg ladder, $S = \frac{1}{2}$	$J_\perp = 4$	$(4, 2)$	3.509×10^{-2}	2.463×10^{-2}	1.1024 ✓L
Ising ladder	$g = 3$	$(4, 3/2)$	1.125×10^{-1}	5.162×10^{-2}	0.6368 ✓L
Mixed-field Ising	$(g, h) = (1/4, 1/20)$	$(6, 97/4)$	9.064×10^{-6}	3.141×10^{-5}	0.3922 ✓L
Heisenberg, $S = 1$	isotropic	$(60, 833/32)$	3.054×10^{-2}	4.796×10^{-2}	0.0729 ✓L

and Weyl's theorems then enclose the sorted exact eigenvalues λ_i of B_q within

$$|\lambda_i - \tilde{\lambda}_i| \leq \frac{2\|G\|_F \max_j |\tilde{\lambda}_j| + (1 + \|G\|_F)\|R\|_F}{1 - \|G\|_F} \qquad (\|G\|_F < 1). \tag{S81}$$

Summing $m_q e^{-\beta\lambda_i}$ over all blocks in interval arithmetic encloses $\log Z_{n_0}(\beta_0)$, $\log Z_{2n_0}(\beta_0)$, and $\log Z_{2n_0}(2\beta_0)$, yielding the certified rational upper bounds (p_0, q_0). The staggered-field, dimerized, ladder, and mixed-field Ising models in Table S5 come from this pipeline directly; the single-ion anisotropy model at $D = 4$ uses the half-length lift below.

When $2n_0$ reaches the limit of block diagonalization, a direct seed at length $2n_0$ is out of reach, since both of its purities need the partition function of a ring of $4n_0$ cells. The spectra already in hand at lengths n_0 and $2n_0$ bound $1 - P_s(n_0, \beta_0) \leq s$, $1 - P_s(n_0, 2\beta_0) \leq s_2$, and $1 - P_\tau(2n_0, \beta_0) \leq t$. The obstacle is the thermal purity at $4n_0$, which no computed spectrum reaches; the torus identity (S22) at length $2n_0$ supplies it:



Move 2 is this identity with the denominator $P_s(2n_0, \beta_0)^2 \leq 1$ dropped ✓L:

$$P_\tau(4n_0, \beta_0) = \frac{P_s(2n_0, 2\beta_0)\, P_\tau(2n_0, \beta_0)^2}{P_s(2n_0, \beta_0)^2} \geq \big(1 - f(s_2)\big)(1 - t)^2. \tag{S82}$$

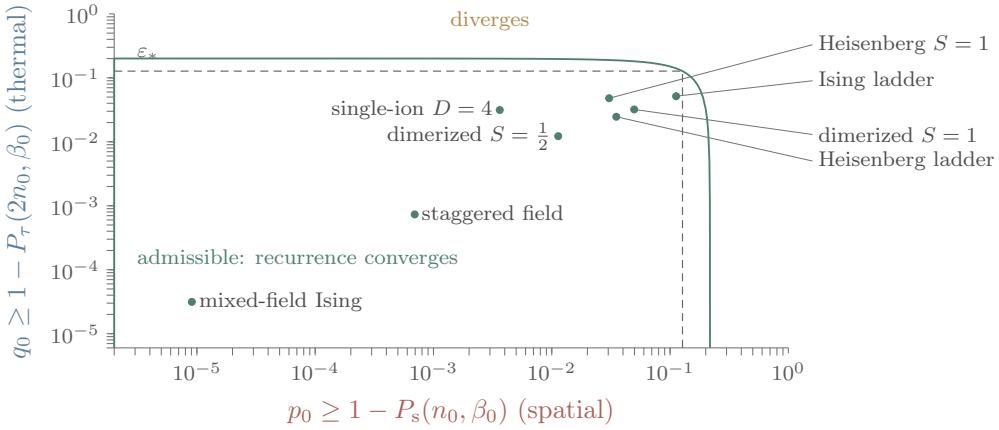
For the spin-1 chain with single-ion anisotropy $D = 4$ at $\beta_0 = 29/10$, the $L = 6$ and $L = 12$ spectra ($3^{12} = 531{,}441$ states) supply s, s_2, and t, and the lift gives the seed at $n_0 = 12$ without the 24-site ring ($3^{24} \simeq 2.8 \times 10^{11}$ states). It sharpens the uniform gap bound at $D = 4$ from 0.6946 (the Lean chain from the direct seed $(6, 9/4)$ ✓L) to 0.8994 for all even $L \geq 12$; the short rings $L = 6, 8, 10$ are covered directly by their verified spectral gaps.

Finally, to verify the J doubling steps (S25) from (p_0, q_0) and the resulting gap bound $\Delta_L > \frac{\log r}{\beta_0}$ in Lean, whose kernel decides rational inequalities but has no certified logarithm, we express the interval bounds (S69) and the quadratic tail of Sec. SI.F in terms of a rational base factor $r > 1$ and tail threshold $\varepsilon \leq 1/100$:

$$r^{2^j} z_j < 1 - z_j \quad (0 \leq j < J), \qquad p_J, q_J \leq \varepsilon \leq \frac{1}{100}, \qquad 5\varepsilon r^{2^J} \leq 1, \tag{S83}$$

where $z_j = 1 - (1 - p_j)^2(1 - q_j)$ along the upward-rounded rational sequence (p_j, q_j). The first inequality is equivalent to $\frac{\log[(1-z_j)/z_j]}{2^j \beta_0} > \frac{\log r}{\beta_0}$ on $[2^j n_0, 2^{j+1} n_0]$, and the remaining two initialize the quadratic tail inequality (S28) for all $j \geq J$. The Lean kernel checks these rational inequalities directly, and the general theorem ✓L concludes that any reflection-positive ring satisfying $1 - P_s(n_0, \beta_0) \leq p_0$ and $1 - P_\tau(2n_0, \beta_0) \leq q_0$ has a unique ground state and $\Delta_L > \frac{\log r}{\beta_0}$

for every even $L \geq n_0$. All eight seeds of Table S5 (labeled by model) lie inside the admissible region of the asymmetric recurrence and below the symmetric threshold $\varepsilon_* \simeq 0.1273$:



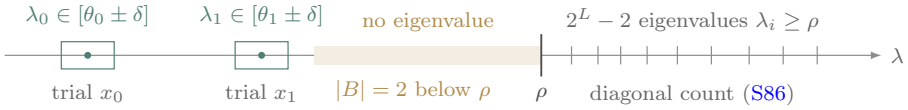
The boundary is the update (S25) iterated in floating point until (p_j, q_j) enters $[0, 1/100]^2$; it admits p_0 up to 0.217 when $q_0 \to 0$ and q_0 up to 0.200 when $p_0 \to 0$, so the recurrence tolerates $1 - P_s$ and $1 - P_\tau$ almost independently.

SIII.C. Unconditional seed proof for the mixed-field Ising chain

Section SIII.B supplies the seed bounds (p_0, q_0) to the Lean theorem as hypotheses. To eliminate that hypothesis and prove an unconditional gap theorem in Lean, the kernel must bound $Z_{n_0}(\beta_0)$, $Z_{2n_0}(\beta_0)$, and $Z_{2n_0}(2\beta_0)$ from above and below directly. A full $2^{12} \times 2^{12}$ eigensolve inside a proof-assistant kernel is prohibitive, but at the mixed-field Ising point $(g, h) = (1/4, 1/20)$,

$$H_L = -\sum_{j=1}^{L} \sigma_j^z \sigma_{j+1}^z - \frac{1}{4} \sum_{j=1}^{L} \sigma_j^x - \frac{1}{20} \sum_{j=1}^{L} \sigma_j^z, \tag{S84}$$

we do not need the full spectrum. On a ring of length $L \in \{6, 12\}$, the longitudinal field splits the two classical ferromagnetic states $|\uparrow \cdots \uparrow\rangle$ and $|\downarrow \cdots \downarrow\rangle$ by $2hL = L/10$, whereas creating a pair of domain walls costs energy 4. Because the transverse field $-g \sum_{j=1}^{L} \sigma_j^x$ has operator norm $gL = L/4$, which at $L \leq 12$ exceeds the ferromagnetic splitting $2hL$ but remains below the domain-wall cost 4, bounding the transverse field by its norm separates all $2^L - 2$ domain-wall states from the two ferromagnetic levels, while two integer trial vectors locate the two lowest eigenvalues to within $\delta = 10^{-10}$:



Two low-lying eigenvalues from integer trial residuals. For any Hermitian matrix H with eigenvalues λ_i, any nonzero trial vector x, and any real shift θ, expanding $\|(H - \theta I)x\|^2$ in the eigenbasis shows that at least one eigenvalue λ_i satisfies $(\lambda_i - \theta)^2 \|x\|^2 \leq \|(H - \theta I)x\|^2$. Because H_L has rational entries in the σ^z basis, this residual inequality reduces to an exact integer dot-product comparison. On each ring $L \in \{6, 12\}$, we supply two integer trial vectors $x_0, x_1 \in \mathbb{Z}^{2^L}$ scaled by 2^{52} and rational candidate energies $\theta_0 < \theta_1$. The Lean kernel checks in exact integer arithmetic that ✓L

$$\|(H_L - \theta_a I)x_a\|^2 \leq \delta^2 \|x_a\|^2 \qquad (a = 0, 1, \quad \delta = 10^{-10}), \tag{S85}$$

which places two distinct eigenvalues of H_L in the disjoint intervals $[\theta_0 - \delta, \theta_0 + \delta]$ and $[\theta_1 - \delta, \theta_1 + \delta]$ ($\theta_0 \simeq -6.3918$ and $\theta_1 \simeq -5.7966$ at $L = 6$; $\theta_0 \simeq -12.7835$ and $\theta_1 \simeq -11.5932$ at $L = 12$).

Lower bound on the remaining $2^L - 2$ eigenvalues by configuration counting. To verify that the two eigenvalues in $[\theta_0 \pm \delta]$ and $[\theta_1 \pm \delta]$ are the two lowest levels of H_L and to bound all remaining $2^L - 2$ eigenvalues from below, we split H_L in the σ^z basis $|\sigma\rangle$ ($\sigma \in \{\pm 1\}^L$) into the diagonal Ising energy $E_{\mathrm{cl}}(\sigma)$ and the off-diagonal transverse

field $-g\sum_{j=1}^{L}\sigma_j^x$. Since each σ_j^x has operator norm one, the transverse field contributes at most $gL\|x\|^2$ to $\langle x, H_L x\rangle$, giving the quadratic-form lower bound

$$\langle x, H_L x\rangle \geq \sum_{\sigma\in\{\pm1\}^L} \big(E_{\rm cl}(\sigma) - gL\big)|x(\sigma)|^2. \tag{S86}$$

For any threshold ρ, let $B = \{\sigma : E_{\rm cl}(\sigma) - gL < \rho\}$. By the min–max principle, at most $|B|$ eigenvalues of H_L lie below ρ, because Eq. (S86) gives $\langle x, H_L x\rangle \geq \rho\|x\|^2$ on the codimension-$|B|$ subspace of vectors vanishing on B. On a ring, the two uniform configurations $\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow$ and $\downarrow\downarrow\downarrow\downarrow\downarrow\downarrow$ have classical energies $-L \mp hL$; every other configuration, such as $\uparrow\uparrow\downarrow\downarrow\downarrow\uparrow\uparrow$, has at least two domain walls ($\sum_j \sigma_j\sigma_{j+1} \leq L-4$) and at least one spin against the field ($\sum_j \sigma_j \leq L-2$), so its shifted classical energy is at least

$$\rho = -(L-4) - h(L-2) - gL, \tag{S87}$$

which equals $-3.7 > \theta_1 + \delta$ at $L = 6$ and $-11.5 > \theta_1 + \delta$ at $L = 12$. The kernel checks $|B| = 2$ at this ρ over all 2^L configurations, so, with two eigenvalues already located below $\theta_1 + \delta < \rho$, the remaining $2^L - 2$ satisfy $\lambda_i \geq \rho$.

Partition-function bounds and unconditional gap theorem. Combining the two eigenvalue enclosures with the lower bound $\lambda_i \geq \rho$ ($i \geq 2$) sandwiches $Z_L(\beta)$ from both sides: dropping all excited states bounds $Z_L(\beta)$ from below by the ground state alone, $Z_L(\beta) \geq e^{-\beta(\theta_0+\delta)}$, while bounding the two lowest eigenvalues from below by $\theta_0 - \delta$ and $\theta_1 - \delta$ and all remaining $2^L - 2$ eigenvalues by ρ gives the upper bound ✓L

$$Z_L(\beta) \leq e^{-\beta(\theta_0-\delta)}\left[1 + e^{-\beta(\theta_1-\theta_0)} + 2^L e^{-\beta(\rho-\theta_0+\delta)}\right]. \tag{S88}$$

The leading factor $e^{-\beta\theta_0}$ cancels between numerator and denominator in both $P_{\rm s}(6,\beta_0) = Z_{12}(\beta_0)/Z_6(\beta_0)^2$ and $P_\tau(12,\beta_0) = Z_{12}(2\beta_0)/Z_{12}(\beta_0)^2$ because $2\theta_0(6) \simeq \theta_0(12)$ to within 10^{-7}. Using rational bounds on e in Lean, evaluating Eq. (S88) at $\beta_0 = 97/4$ proves the starting bounds at $(n_0,\beta_0) = (6, 97/4)$ ✓L,

$$p_0 = \frac{9064}{10^9} \geq 1 - P_{\rm s}(6, 97/4), \qquad q_0 = \frac{3141}{10^8} \geq 1 - P_\tau(12, 97/4). \tag{S89}$$

At $j = 0$, these bounds give $z_0 = 1-(1-p_0)^2(1-q_0) \simeq 4.954\times10^{-5}$, so $(1-z_0)/z_0 > 20{,}184$; substituting Eq. (S89) into the rational doubling inequalities (S83) over $J = 14$ steps with base factor $r = 13531935/10^3 < (1-z_0)/z_0$ proves a unique ground state and ✓L

$$\boxed{\Delta_L > \frac{\log(13531.935)}{97/4} > 0.3922 \qquad \text{for every even } L \geq 6} \tag{S90}$$

unconditionally in Lean.

SIV: Lean 4 formalization

A computer-assisted proof combines an analytic reduction with a finite numerical calculation. Formalizing the periodic and open-chain spin-1 proofs of Refs. [10, 21] directly in Lean 4 [22] faces three obstacles:

1. *Infinite-dimensional path space.* The continuous-time spatial transfer kernel of Ref. [10] acts on L^2 of Poisson bond histories (Sec. SV.B); formalizing its trace identities directly would require path-space measure and operator machinery not yet in Mathlib.

2. *Model specificity.* Writing the analytic steps around the isotropic spin-1 Heisenberg Hamiltonian ties the formalization to one model, so none of the operator inequalities can be reused for other spin chains or ladders.

3. *Finite numerical inputs.* The seed bounds rely on external computations—block diagonalizations, short-ring trace recursions, and matrix-product trial contractions—whose floating-point or C/Python implementations cannot be trusted inside a proof assistant.

The diagrammatic reformulation in the main text and Secs. SI–SIII (and Sec. SV for open chains) resolves all three obstacles before formalization begins. First, discretizing imaginary time into M brickwork slices replaces the path-space kernel by the finite-dimensional Gram matrix $T_{\beta,M} = \mathcal{F}^{\mathsf{T}}\mathcal{F} \succeq 0$ [Eq. (S9)]. Instead of constructing a limiting

continuous-time transfer operator in Lean, we prove the three closed spatial power-sum inequalities (S16) at finite M and pass them to the exact partition functions $Z_L(\beta)$ via the fixed-(L,β) matrix product limit of Sec. SI.C ✓L. Likewise, to state ground-state uniqueness and $\Delta_L = E_1(L) - E_0(L) > c$ without sorting the eigenvalues of H_L, we require a single ground-state index m_0 separated from every other level $m \neq m_0$ by $E_L(m) - E_L(m_0) > c$ ✓L. Second, isolating the bond reflection-positivity identity $-h = \sum_\alpha C_\alpha \otimes FC_\alpha F$ with $FoF = o$ [Eq. (S71)] makes the periodic scale-doubling theorem apply uniformly to any reflection-positive unit-cell ring ✓L.

TABLE S6. Structure of the Lean 4 formalization [23] and its interactive proof blueprint [24]. Each row lists a chapter of the blueprint, the corresponding manuscript sections, the theorem proved in Lean 4, and the finite numerical hypotheses (if any) on which that theorem depends.

Blueprint chapter	Manuscript	Theorem proved in Lean 4	Numerical hypotheses
2. Two-purity criterion	Main text; Sec. SI.F	Uniform gap from purities $\geq 1 - \varepsilon_0$ ✓L	None
3. Asymmetric doubling & chains	Secs. SI.D, SIII.B	Uniform gap from rational doubling chain ✓L	None (kernel-checked)
4. Reflection-positive rings	Secs. SI.A–SI.C, SIII.A	Reflection-positive ring gap theorem ✓L	None
5. Trace bounds & sector tables	Sec. SII	Seed purities from short-ring moments ✓L	None
6. Spin-1 Heisenberg ring	Main text; Sec. SII	Spin-1 ring gap $\Delta_L > 0.07290$ $(L \geq 36)$ ✓L	Short-ring traces $(4 \leq n \leq 12)$
7. Open chains with end fields	Sec. SV	Open spin-1 gap $(n \geq 1921)$ ✓L	Four seeds at $\beta_0 = 49$
8. Spin chains & ladders	Sec. SIII; Table S5	Uniform gaps for Table I models ✓L	Seed pair (p_0, q_0)
9. Kernel-checked seeds	Secs. SII, SIII.C	Unconditional Ising gap; $S = 1$ trials ✓L	None (Ising); $n \geq 5$ traces

Third, separating the universal doubling recurrence from the finite seed allows the formalization to treat the numerical inputs at two levels (Table S6):

1. *Seed-hypothesis theorems (Chapters 6–8).* Because the asymmetric update (S25) is a rational map and the interval condition (S83) replaces logarithms by rational power comparisons $r^{2^j} z_j < 1 - z_j$, the step from any rational seed pair (p_0, q_0) to a uniform gap $\Delta_L > (\log r)/\beta_0$ reduces to finitely many rational inequalities checked in the Lean kernel. Taking (p_0, q_0) (or the short-ring trace table of Table S2) as an explicit hypothesis instantiates machine-checked gap theorems for all eight periodic models in Table I and for the open spin-1 chain with endpoint fields (Theorem S1), while allowing the seed enclosures to be certified externally by verified block diagonalizations (Sec. SIII.B) or exact integer trace recursions (Sec. SII).

2. *Kernel-checked seed arithmetic (Chapter 9).* Where the finite seed calculation can be reduced to exact integer identities, the Lean kernel verifies the seed calculation itself. For the mixed-field Ising chain at $(g, h) = (1/4, 1/20)$ (Sec. SIII.C), the kernel checks the integer trial-vector residuals (S85) and the classical configuration count (S87), proving the uniform gap (S90) unconditionally without external numerical hypotheses ✓L. For the periodic and open spin-1 chains, the kernel checks the exact integer contractions (S45) of the 72- and 120-site periodic matrix-product trial states, the four open-chain trial states on 34, 78, 142, and 240 sites, and the shortest ring trace recurrences ($n = 4$ periodic and nine of fifteen boundary traces), reducing the remaining hypotheses to the short-ring integer trace totals for $5 \leq n \leq 12$ ✓L and the remaining six boundary traces ✓L.

The human-readable proof blueprint [24] (`https://sirui-lu.com/CertifiedGaps/blueprint/`) links every definition, lemma, and theorem in the dependency graph to its Lean 4 declaration, and the companion repository [23] bundles the manuscript, standalone numerical certificate checkers, and formal proofs following the Agentic Publication Protocol [35].

SV: Open spin-1 chains with boundary fields

SV.A. Open-chain gap theorem and boundary remainder

We extend the diagrammatic doubling argument from periodic rings to open chains with endpoint magnetic fields. Consider the spin-1 Heisenberg chain with equal endpoint fields ($\uparrow\cdots\uparrow$),

$$H_n^{\mathrm{s}} = \sum_{i=1}^{n-1} \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} - h(S_1^z + S_n^z), \qquad h = \frac{3}{5}, \tag{S91}$$

where $\sum_\alpha (S^\alpha)^2 = 2I$ and the exchange coupling is one. We give a diagrammatic proof of the following theorem from Ref. [21], using its certified finite inputs.

Theorem S1 (Boundary-field gap [21] ✓L)**.** *For every odd $n \geq 1921$, H_n^{s} has a unique ground state and*

$$\Delta_n > \frac{\log 10}{392} \simeq 0.005874.$$

The two constants come from three doublings of the certified seed $(M_2, \beta_2) = (240, 49)$; scale 5 is the first at which every tracked bound is below 10^{-2}:

$$M_5 = 2^3 \cdot 240 = 1920, \qquad \beta_5 = 2^3 \cdot 49 = 392, \qquad n \geq M_5 + 1 = 1921.$$

Without the endpoint fields, the effective spin-1/2 modes at the two ends produce four low-lying states whose splitting is exponentially small in the chain length [43]. Applying equal fields on an odd chain lifts this boundary degeneracy [21].

Our strategy is to bound the thermal purity of the open chain at a scale β proportional to the length, just as in the periodic argument. Order the Gibbs weights of H_n^{s} as $w_0 \geq w_1 \geq \cdots$. The thermal purity $P_\tau^{\mathrm{s}}(n,\beta) = \sum_m w_m^2$ then bounds the leading weight and the gap:

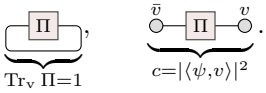
$$w_0 \geq P_\tau^{\mathrm{s}}(n,\beta), \qquad e^{-\beta\Delta_n} \leq \frac{1 - P_\tau^{\mathrm{s}}(n,\beta)}{P_\tau^{\mathrm{s}}(n,\beta)}, \qquad \Delta_n = E_1(n) - E_0(n). \tag{S92}$$

A thermal purity above 1/2 forces a unique ground state and bounds the gap from below. What changes in the open-chain setting is how we propagate the thermal purity across doubled lengths and inverse temperatures.

To see where the new difficulty enters, consider the spatial transfer operator at a fixed temperature. Suppose that X_β has a simple positive leading eigenvalue λ (▮), and write the normalized transfer operator as

$$T := X_\beta/\lambda = \Pi + R, \qquad \Pi = |\psi\rangle\langle\psi|, \qquad \Pi R = R\Pi = 0, \quad \|\psi\| = 1. \tag{S93}$$

In a periodic ring, the trace weights the leading projector Π by one. In an open chain, the chain ends are contracted with boundary vectors, so the leading contribution is weighted instead by how much of v lies along ψ (∠):



(S94)

Because c is not fixed by normalization and could a priori vanish, squaring in space no longer closes on its own. With T self-adjoint, the same boundary vector at both ends, and an even length m, the remainder of the normalized boundary matrix element is a squared norm, hence nonnegative:

$$\begin{aligned} B_m &= \bar{v} \text{—} T^m \text{—} v \\ &= \bar{v} \text{—} \Pi \text{—} v + \bar{v} \text{—} R^m \text{—} v = c + D_m, \\ D_m &= \|R^{m/2} v\|^2 \geq 0. \end{aligned} \tag{S95}$$

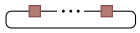
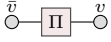
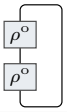
To show that the leading eigenvector dominates B_m, we bound the remainder's share of B_m rather than the remainder itself,

$$e_m := \frac{D_m}{B_m} = 1 - \frac{c}{B_m}, \tag{S96}$$

the grey fraction of ▬ at the certified seed value $e_{142} \leq .14$ of Eq. (S105). Increasing the chain length suppresses D_m by powers of $\|R\|$ while leaving c unchanged.

Doubling the inverse temperature changes both v and ψ, so c does not bound the new overlap c'; Step 1 (Sec. SV.D.a) removes c' by comparing two lengths at the new temperature.

The recurrence therefore tracks three contractions at every scale j, each with its small quantity and its certified seed value:

	contraction	tracked quantity	seed ($j=2$)	controlled in
(R)	 ring purity	$\varepsilon_j \geq 1-P_{\mathrm{s}},\ 1-P_\tau$	.052	Sec. SV.C
(E)	 boundary overlap	$e_{k_j}(j) = D/B$	.14	Sec. SV.D.a
(T)	 opposite-field purity	$1-P_\tau^{\mathrm{o}}(M_j,\beta_j)$	.21	Secs. SV.D.b–SV.D.c

Section SV.B constructs the boundary vectors, Sec. SV.E runs three explicit steps from $j=2$ to $j=5$ and closes the induction, and Sec. SV.F passes to equal fields and odd lengths.

SV.B. Continuous-time transfer kernel and boundary vectors

We first represent the open-chain partition functions as transfer-operator matrix elements using the continuous-time bond expansion of Ref. [21], which sums over the times of bond insertions rather than placing them on a finite Trotter grid. As in the periodic proof (Sec. SII.A.a), we shift the Hamiltonian by a constant κ per site. The shift is chosen so that $-\kappa$ lies just above the energy per site of the 72-site matrix-product trial state of Ref. [10] ($E_0(72)/72 < -\kappa$, Sec. SII.A.b):

$$\kappa = \frac{700741}{500000} = 1.401482. \tag{S97}$$

Throughout this section, Z_n and all open-chain partition functions include the shift $\kappa n I$. It places the leading transfer eigenvalue near one at the seed scales and cancels from every purity and spectral gap.

In the Cartesian basis, $G_\alpha = iS^\alpha$ is real antisymmetric and

$$-\boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} = \sum_{\alpha=x,y,z} G_{\alpha,i} G_{\alpha,i+1}.$$

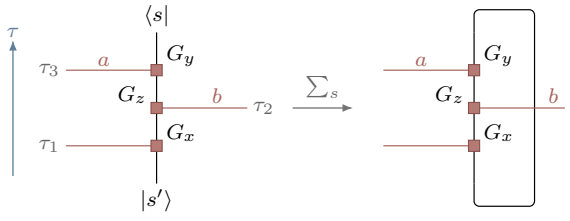
Expanding $e^{-\beta H}$ on each bond produces a bond history a, an ordered list of events (τ,α) with $0<\tau<\beta$ and $\alpha \in \{x,y,z\}$. The bond measure μ_β sums the labels and integrates the ordered times, with unit mass on the empty history:

$$\mu_\beta(\text{all histories}) = \sum_{k\geq 0} \underbrace{3^k}_{k \text{ labels}} \underbrace{\frac{\beta^k}{k!}}_{0<\tau_1<\cdots<\tau_k<\beta} = e^{3\beta}.$$

Between the left bond history a and the right bond history b of a site, the shifted local tensor is

$$W_\beta^{ss'}(a,b) = e^{-\kappa\beta} \langle s| \mathcal{T}_\tau \prod_{(\tau,\alpha)\in a\cup b} G_\alpha |s'\rangle, \tag{S98}$$

where $\mathcal{T}_\tau$ orders the events of both bonds by time, later ones to the left:



$W_\beta^{ss'}(a,b) = \langle s|G_yG_zG_x|s'\rangle$ $\qquad$ $X_\beta(a,b) = \mathrm{Tr}(G_yG_zG_x)$

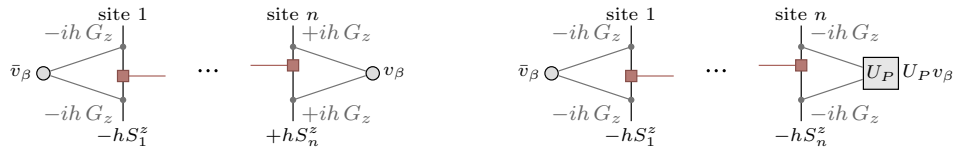
Tracing the physical indices defines the transfer kernel $X_\beta(a,b)$ on $L^2(\mu_\beta)$. It is real, and symmetric because the product follows the times, not the sides.

To represent a boundary field along z, we contract the endpoint bond history with the boundary vector

$$v_\beta(a) = \begin{cases} (ih)^{|a|}, & \text{all event labels are } z, \\ 0, & \text{otherwise,} \end{cases} \qquad \|v_\beta\|^2 = e^{\beta h^2}. \tag{S99}$$

Its value on the empty history is one, and its norm follows by summing $(\beta h^2)^k/k!$ over the number k of z events.

Each event on the boundary bond inserts $G_z = iS^z$ at the end site at its time. It enters the time-ordered product like an interior bond event, and summing counts and integrating times exponentiates the endpoint field. Conjugation reverses the sign of ih, so contracting with v_β at both ends gives the auxiliary opposite-field chain $\uparrow\bullet\bullet\bullet\bullet\downarrow$ (left). The π rotation $P = \mathrm{diag}(1,-1,-1)$ about x acts on the right boundary vector as U_P; it negates every z event there and restores equal fields $\uparrow\bullet\bullet\bullet\bullet\uparrow$ (right):



(S100)

$$H_n^{\mathrm{o}} = H_n^{\mathrm{s}} + 2hS_n^z. \tag{S101}$$

The ring bounds of Sec. SV.C concern the spectrum of X_β, so both open-chain partition functions must become matrix elements of it. In diagram form,

$$e^{-\beta(H_n^{\mathrm{o}}+\kappa nI)} = \quad \bar v_\beta \text{—} W_\beta \text{—} \cdots \text{—} W_\beta \text{—} v_\beta ,$$

$$Y_n(\beta) := \mathrm{Tr}\, e^{-\beta(H_n^{\mathrm{o}}+\kappa nI)} = \quad \bar v_\beta \text{—} X_\beta^n \text{—} v_\beta ,$$

$$Y_n^{\mathrm{s}}(\beta) := \mathrm{Tr}\, e^{-\beta(H_n^{\mathrm{s}}+\kappa nI)} = \quad \bar v_\beta \text{—} X_\beta^n \text{—} U_P v_\beta . \tag{S102}$$

The two end bond histories meet the boundary vectors, and the physical trace at every site turns the row of tensors into X_β^n; the two chains differ only by U_P at the right end.

Spin rotations act on bond histories by real orthogonal operators that commute with X_β, so $[U_P, X_\beta] = 0$. Contracting the time-ordered expansion site by site is exact by Fubini's theorem, since $\|G_\alpha\| = 1$ bounds the absolutely summed series by $3^n e^{3\beta(n-1)+2\beta h}$ before the energy shift.

The recurrence runs on the auxiliary opposite-field chain at even length m, not on the odd equal-field chain of the theorem. With the same boundary vector at both ends, the remainder of $Y_m(\beta)$ is the squared norm $D_m \geq 0$ of Eq. (S95), on which the doubling closes.

SV.C. Bulk ring purities and transfer-operator bounds

The open-chain recurrence of Steps 1–3 needs three facts about the bulk transfer operator at every scale j. They are a simple positive leading eigenvalue that U_P fixes, a decay rate for the remainder across half a chain, and the ratio r_j of leading eigenvalues at consecutive temperatures. None is certified directly at large j, but the periodic argument proves ring purity bounds at every scale from one certified seed, and this subsection extracts the three facts from those purities. The open chain thus borrows its bulk control from the ring; only the boundary needs new inputs.

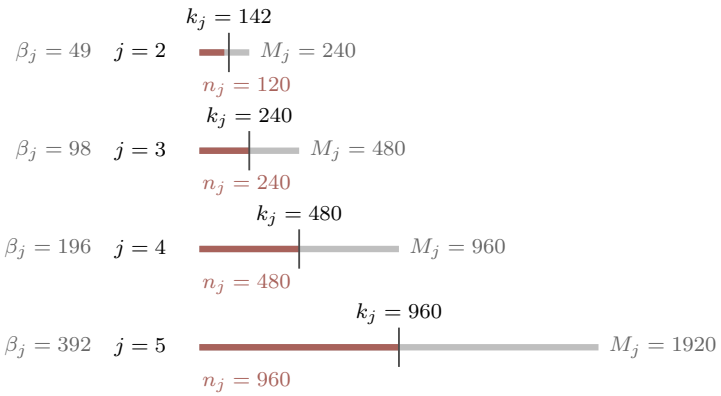
The recurrence runs over the dyadic scales

$$\beta_j = 49\,2^{j-2}, \qquad n_j = 120\,2^{j-2}, \qquad M_j = 2n_j, \qquad j \geq 2, \tag{S103}$$

starting at the certified point $j = 2$ of Ref. [21]. The ring length controls the bulk and the open length carries the thermal purity. The boundary ratio e_k is tracked at the certified length 142 first and at the previous open length thereafter,

$$k_2 = 142, \qquad k_{j+1} = M_j, \tag{S104}$$

so $k_j = M_j/2$ for $j \geq 3$:



Ref. [21] certifies four inputs at $j = 2$ ✓L:

input	controls	weights	bound
$P_{\mathrm{s}}(120, 49) \geq .948$	bulk spectrum		$\varepsilon_2 = 1 - .948 = .052$
$P_\tau(240, 49) \geq .948$	bulk spectrum		ε_2
$e_{142}(2) \leq .14$	boundary overlap		$E_2 = .14$
$P^{\mathrm{o}}_\tau(240, 49) \geq .79$	thermal		$\eta_2 = 1 - .79 = .21$

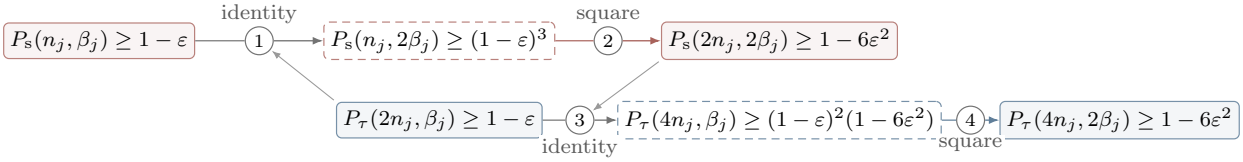
(S105)

Here $P_{\mathrm{s}}(n, \beta) = Z_{2n}(\beta)/Z_n(\beta)^2$ and $P_\tau(n, \beta) = Z_n(2\beta)/Z_n(\beta)^2$ are ring purities, $e_m(j)$ is Eq. (S96) at β_j, and $P^{\mathrm{o}}_\tau(m, \beta) = Y_m(2\beta)/Y_m(\beta)^2$.

The periodic argument squares $1 - P_{\mathrm{s}}$ and $1 - P_\tau$ with constant 5 only below the threshold $1/100$, which the seed $\varepsilon_2 = .052$ exceeds; repeating the four-step update with constant 6 gives

$$P_{\mathrm{s}}(n_j, \beta_j),\ P_\tau(M_j, \beta_j) \geq 1 - \varepsilon_j, \qquad \varepsilon_2 = .052, \qquad \varepsilon_{j+1} = 6\varepsilon_j^2. \tag{S106}$$

One scale of the periodic argument, with $\varepsilon = \varepsilon_j$, updates the spatial and thermal purities in four moves:



Squaring and renormalizing a positive trace-one operator of purity P concentrates its weights (↦), and its new purity P' satisfies

$$1 - P' \leq f(1 - P), \qquad f(z) = \frac{z^2}{2(1 - z)^2}. \tag{S107}$$

For $0 \leq \varepsilon \leq .052$, Move 2 gives

$$f(1 - (1 - \varepsilon)^3) = \frac{\varepsilon^2}{2}\left[\sum_{s=1}^{3}(1 - \varepsilon)^{-s}\right]^2 \leq 6\varepsilon^2 \leq \varepsilon, \tag{S108}$$

the bracket being at most 3.35 at $\varepsilon = .052$, so the prefactor is at most $5.6 < 6$. Since $(1 - \varepsilon)^2(1 - 6\varepsilon^2) \geq (1 - \varepsilon)^3$, Move 4 gives the same bound $6\varepsilon^2$, and induction on $j \geq 2$ proves Eq. (S106).

Write λ_j for the leading eigenvalue of X_{β_j} and $T_j = X_{\beta_j}/\lambda_j = \Pi_j + R_j$ as in Eq. (S93). The three bulk facts, each with the step that consumes it, are ✓L

(i) $\lambda_j > 0$ simple, $U_P \Pi_j = \Pi_j$ — overlap c in Step 1 and Sec. SV.F,

(ii) $\|R_j\|^{n_j} \leq d(\varepsilon_j) < 1$, $d(\varepsilon) = \dfrac{\varepsilon}{2 - 3\varepsilon}$ — suppression factor q in Step 1,

(iii) $r_j^{M_j} \geq (1 - \varepsilon_j)(1 - \varepsilon_{j+1})$, $r_j = \dfrac{\lambda_{j+1}}{\lambda_j^2} \leq 1$ — bulk factor in Step 2 and Sec. SV.F. (S109)

No transfer eigenvalue is computed, so (i) and (ii) must follow from the ring purity alone. A purity at least $1-\varepsilon > 1/2$ forces one weight p_0 above $1/2$ (), and its tail bounds every subleading ratio. The purity is at most p_0, so the tail $z = 1 - p_0$ is below $1/2$. The purity is also at most $p_0^2 + z^2 = 1 - 2z(1-z)$, so $2z(1-z) \le \varepsilon$ and

$$z \le \varepsilon, \qquad z \le \frac{\varepsilon}{2(1-\varepsilon)}, \qquad \frac{z}{1-z} \le \frac{\varepsilon}{2(1-\varepsilon)-\varepsilon} = d(\varepsilon).$$

Applying this estimate to the normalized weights $p_i = |\lambda_i(\beta_j)|^{n_j}/Z_{n_j}(\beta_j)$ yields the subleading tail bound

$$\sum_{i\ge 1} \left| \frac{\lambda_i(\beta_j)}{\lambda_j} \right|^{n_j} \le d(\varepsilon_j). \tag{S110}$$

Since $d(\varepsilon_j) < 1$, three consequences follow.

- λ_j is simple, and $\|R_j\|^{n_j} \le d(\varepsilon_j) < 1$ is property (ii).
- $\lambda_j > 0$. A negative leading eigenvalue would make $\mathrm{Tr_v}\, X_{\beta_j}^N < 0$ for all large odd N, because X_{β_j} is Hilbert–Schmidt and the rest of its spectrum has smaller radius; but $Z_N(\beta_j) > 0$.
- $U_P \Pi_j = \Pi_j$. The cyclic axis permutation and the three π rotations commute with the real kernel X_{β_j}, so each acts by a sign on the leading eigenline. The permutation has order three, so its sign is $+1$. It conjugates the three rotations into one another, so their signs agree; two of them multiply to the third, so the common sign is $+1$.

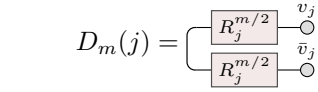
Property (iii) bounds the bulk factor that Step 2 loses when the temperature doubles. The two transfer operators do not commute, so we compare both leading eigenvalues with one ring partition function. First, $P_\tau(N, \beta_j) \le 1$ and even Nth roots as $N \to \infty$ give $\lambda_{j+1} \le \lambda_j^2$, so $r_j \le 1$. At length $M_j = n_{j+1}$,

$$\begin{aligned}
\lambda_{j+1}^{M_j} &\ge (1-\varepsilon_{j+1}) Z_{M_j}(\beta_{j+1}) && \text{leading weight at } \beta_{j+1},\\
&\ge (1-\varepsilon_{j+1})(1-\varepsilon_j) Z_{M_j}(\beta_j)^2 && \text{ring thermal purity at } \beta_j,\\
&\ge (1-\varepsilon_{j+1})(1-\varepsilon_j) \lambda_j^{2M_j} && \text{even } M_j\text{: trace} \ge \text{leading term},
\end{aligned} \tag{S111}$$

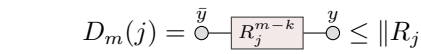
and dividing by $\lambda_j^{2M_j}$ proves (iii). The bulk is now controlled at every scale; the rest of this section concerns the boundary only.

SV.D. One doubling step for the opposite-field chain

Normalizing by λ_j^m removes the extensive part of the partition function, so the recurrence tracks only the overlap and the remainder. At scale j, write $v_j = v_{\beta_j}$ and, for even $m \ge 2$, define the normalized opposite-field partition function, its remainder, and the boundary remainder ratio by

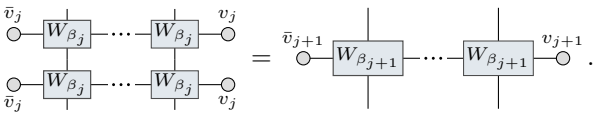
$$\begin{aligned}
c_j &= |\langle \psi_j, v_j \rangle|^2,\\
B_m(j) &= \frac{Y_m(\beta_j)}{\lambda_j^m} = c_j + D_m(j) > 0,\\
D_m(j) &= \text{  } = \|R_j^{m/2} v_j\|^2, \qquad e_m(j) = \frac{D_m(j)}{B_m(j)}.
\end{aligned} \tag{S112}$$

Folding only $R_j^{k/2}$, $k \le m$ even, onto each boundary vector leaves R_j^{m-k} in the middle, and the remainder decays geometrically with the extra length ():

$$D_m(j) = \text{  } \le \|R_j\|^{m-k} D_k(j), \qquad y = R_j^{k/2} v_j, \quad D_m(j) \ge 0. \tag{S113}$$

In particular $c_j \le B_m(j) \le B_k(j)$; Steps 1 and 2 use this monotonicity in the length.

Steps 1 and 2 also need the normalized contraction at the doubled temperature in terms of quantities at β_j. Two rows of Eq. (S102) stacked in imaginary time do not fuse site by site, but the full operators multiply:



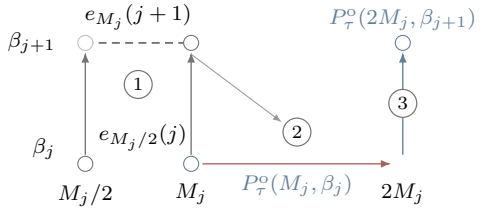
(S114)

Its trace, normalized by $Y_m(\beta_j)^2$, is the thermal purity $P^{\circ}_{\tau}(m,\beta_j) = \operatorname{Tr}\rho^{\circ}_m(\beta_j)^2 \in (0,1]$ of the Gibbs state $\rho^{\circ}_m(\beta) = e^{-\beta(H^{\circ}_m+\kappa m I)}/Y_m(\beta)$. Substituting $Y_m(\beta_j) = \lambda_j^m B_m(j)$ and $\lambda_{j+1} = r_j\lambda_j^2$ relates the normalized contractions at the two temperatures:

$$B_m(j+1) = P^{\circ}_{\tau}(m,\beta_j)\, r_j^{-m} B_m(j)^2. \tag{S115}$$

The thermal purity is the only factor not yet tracked; it is entry (T) of the recurrence.

Given the boundary remainder ratio at length $M_j/2$ and the thermal purity at length M_j, both at β_j, one step to scale $j+1$ moves through the (length, inverse temperature) plane in a fixed order; the key lists the three bounds that Steps 1–3 below derive:



① $e_{M_j}(j+1) \le \dfrac{q}{1-q}\Big[\dfrac{1}{(1-e_{M_j/2}(j))^2\, P^{\circ}_{\tau}(M_j,\beta_j)} - 1\Big], \quad q \ge \|R_{j+1}\|^{M_j/2}$

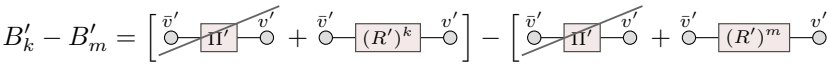
② $P^{\circ}_{\tau}(2M_j,\beta_j) \ge r_j^{M_j}\,[1-e_{M_j}(j+1)]\,P^{\circ}_{\tau}(M_j,\beta_j)$

③ $1 - P^{\circ}_{\tau}(2M_j,\beta_{j+1}) \le f\big(1-P^{\circ}_{\tau}(2M_j,\beta_j)\big)$

The proved pair sits at $k_{j+1} = M_j$ and $M_{j+1} = 2M_j$, so the step repeats.

a. Step 1: Boundary remainder at the doubled inverse temperature

Step 1 must bound the boundary remainder ratio at the doubled temperature, which Step 2 then uses. The old overlap says nothing about the new one c'. But c' does not depend on the length, so it cancels from the difference of two lengths at the new temperature, and Eq. (S115) returns their ratio to scale j. Fix even lengths $m > k \ge 2$, assume $e_k(j) < 1$, and write primes for scale $j+1$. Choose $q < 1$ with $\|R_{j+1}\|^{m-k} \le q$, so that Eq. (S113) gives $D'_m \le qD'_k$. Subtracting the two lengths cancels c':



(S116)

Combining $D'_k = D'_m + B'_k - B'_m$ with $D'_m \le qD'_k$ gives

$$(1-q)D'_m \le q(B'_k - B'_m), \qquad e'_m \le \frac{q}{1-q}\left(\frac{B'_k}{B'_m} - 1\right). \tag{S117}$$

Equation (S115) at both lengths expresses the new-temperature ratio in terms of quantities at scale j:

$$\begin{aligned}\frac{B'_k}{B'_m} &= \underbrace{\frac{P^{\circ}_{\tau}(k,\beta_j)}{P^{\circ}_{\tau}(m,\beta_j)}}_{\le 1/P^{\circ}_{\tau}(m,\beta_j)}\ \underbrace{r_j^{m-k}}_{\le 1}\ \underbrace{\left[\frac{B_k(j)}{B_m(j)}\right]^2}_{\le [1-e_k(j)]^{-2}} \\ &\le \frac{1}{P^{\circ}_{\tau}(m,\beta_j)[1-e_k(j)]^2},\end{aligned} \tag{S118}$$

the last since $B_m(j) \ge c_j = [1-e_k(j)]B_k(j) > 0$. Once $e'_m < 1$ follows, so does $c' = (1-e'_m)B'_m > 0$.

Equation (S117) is a product of two small factors, the decay $q/(1-q)$ over the length difference and the excess of B'_k/B'_m above one, which is why the boundary remainder ratio squares. For the induction, both factors must be $O(x)$ once every tracked bound is at most x. Take $m = M_j$ and $k = M_j/2$ with $\varepsilon_j \le x \le .01$. Property (ii) at scale $j+1$ gives $\|R_{j+1}\|^{M_j} \le d(\varepsilon_{j+1}) \le d(6x^2)$, so across the half-chain $m-k = M_j/2$ we may take

$$q = \sqrt{d(\varepsilon_{j+1})} \le \sqrt{\frac{6x^2}{2-18x^2}} \le 2x. \tag{S119}$$

Separately, when $e_k(j) \le x$ and $P^{\mathrm{o}}_\tau(m,\beta_j) \ge 1-x$, Eq. (S118) bounds the excess above one by

$$\frac{B'_k}{B'_m} - 1 \le (1-x)^{-3} - 1 = O(x). \tag{S120}$$

Multiplying Eqs. (S119) and (S120) in Eq. (S117) bounds the new boundary remainder ratio e'_m by $O(x^2)$.

b. *Step 2: Thermal purity at the doubled length*

Step 2 must bound the thermal purity at length $2M$ and the old temperature β_j, which Step 3 squares. Doubling an open chain is not a squaring, since the boundary vectors appear once, so no purity inequality applies directly. Instead, for even M, dividing Eq. (S115) at length $2M$ by that at length M gives three factors, each already controlled:

$$\begin{aligned}\frac{P^{\mathrm{o}}_\tau(2M,\beta_j)}{P^{\mathrm{o}}_\tau(M,\beta_j)} &= \underbrace{r_j^M}_{\text{(iii)}} \underbrace{\frac{B_{2M}(j+1)}{B_M(j+1)}}_{\ge 1-e_M(j+1)\ \text{(Step 1)}} \underbrace{\left[\frac{B_M(j)}{B_{2M}(j)}\right]^2}_{\ge 1\ \text{(S113)}} \\ &\ge r_j^M[1-e_M(j+1)].\end{aligned} \tag{S121}$$

The doubling thus costs only the bulk factor of (iii) and the Step 1 ratio.

c. *Step 3: Thermal squaring at the doubled length*

Step 3 returns to the dyadic grid at $(2M_j,\beta_{j+1})$, where the next scale starts. The open-chain Gibbs state is positive with unit trace, so the periodic squaring inequality applies to it unchanged; the boundary plays no role. Squaring the Gibbs state $\rho^{\mathrm{o}}_{2M}(\beta_j)$, Eq. (S114), and applying the squaring inequality (S107) doubles the inverse temperature at length $2M$:

$$1 - P^{\mathrm{o}}_\tau(2M,\beta_{j+1}) \le f(1-P^{\mathrm{o}}_\tau(2M,\beta_j)). \tag{S122}$$

With Steps 1 and 2 this closes one scale of the recurrence.

SV.E. Initial rational steps and the small-remainder induction

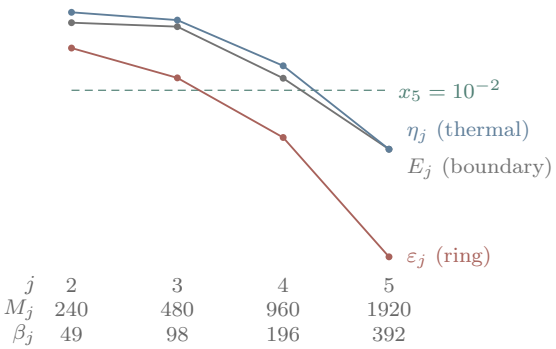
The three steps of Sec. SV.D now become a recurrence across $j \ge 2$ on two numbers per scale, iterated from the seeds, with the comparison lengths k_j of Eq. (S104). Let E_j and η_j denote upper bounds on $e_{k_j}(j)$ and $1-P^{\mathrm{o}}_\tau(M_j,\beta_j)$. Step 1 needs the decay over the length difference, while (ii) bounds it over the ring length; a fractional power converts one to the other. One scale $j \to j+1$ updates them by ✓L

$$\begin{aligned}
F(q,e,t) &= \frac{q}{1-q}\left[\frac{1}{(1-e)^2 t} - 1\right], && \text{Step 1, Eqs. (S117)–(S118),} \\
q_j &= d(\varepsilon_{j+1})^{(M_j-k_j)/M_j}, && \text{(ii) at scale } j+1, \text{ where } n_{j+1}=M_j, \\
E_{j+1} &\ge F(q_j,E_j,1-\eta_j), && \text{Step 1,} \\
A_j &= (1-\eta_j)(1-\varepsilon_j)(1-\varepsilon_{j+1})(1-E_{j+1}), && \text{Step 2, Eq. (S121): } P^{\mathrm{o}}_\tau(2M_j,\beta_j) \ge A_j, \\
\eta_{j+1} &\ge f(1-A_j), && \text{Step 3, Eq. (S122),}
\end{aligned} \tag{S123}$$

valid whenever all four bounds lie in $[0,1)$; F increases in q and e and decreases in t, so any upper bound on q_j may be substituted.

The seed bounds $E_2 = .14$ and $\eta_2 = .21$ are too large for the analytical induction below, which needs $x \le .01$, so we first run Eq. (S123) for three explicit rational steps:

j	E_j	η_j	$(M_j - k_j)/M_j$	$q_j \le$
2	.14	.21	49/120	.142
3	.12	.155	1/2	.0282
4	.016	.026	1/2	.0028
5	.001	.001	—	—



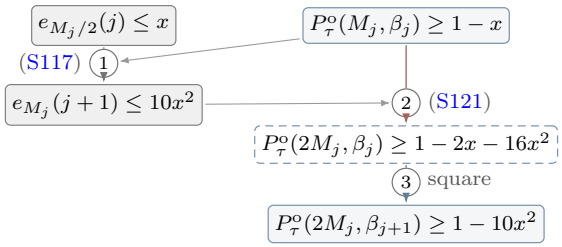
Following Ref. [21], each entry is rounded upward to a simple rational. The fractional powers then reduce to integer power comparisons that a proof assistant checks directly, $d(\varepsilon_3)^{49} < .142^{120}$ for q_2. Each row follows from the previous one; at $j = 2$,

$$E_3: \ \frac{q_2}{1-q_2} \simeq .166, \quad \frac{1}{(1-E_2)^2(1-\eta_2)} - 1 \simeq .712, \quad .166 \cdot .712 \simeq .118 \le .12,$$
$$\eta_3: \ A_2 = (1-\eta_2)(1-\varepsilon_2)(1-\varepsilon_3)(1-E_3) \simeq .79 \cdot .948 \cdot .984 \cdot .88 = .648, \quad f(.352) \simeq .147 \le .155.$$

The first step gains little because q_2 is still large; once q_j is small, the bounds square, as the plot above shows. With $\varepsilon_5 = 6\varepsilon_4^2 < .000015$, the $j = 5$ row brings all three bounds below .01, the base case of the small-remainder induction:

$$\varepsilon_j, \ e_{k_j}(j), \ 1 - P_\tau^o(M_j, \beta_j) \le x_j, \quad j \ge 5; \qquad x_5 = .01, \quad x_{j+1} = 10x_j^2, \quad x_j = 10^{-1-2^{j-5}}. \tag{S124}$$

The induction step of Eq. (S124) must return all three bounds from x to $10x^2$ in one scale. Each step is quadratic in x, but Step 2 consumes the Step 1 output and the ring bound ε_j enters both q and A, so the constants are tracked jointly. Assume that the three bounds hold at scale $j \ge 5$ and write $x = x_j \le .01$. One step from scale j to scale $j+1$ ($M_{j+1} = 2M_j$) updates the boundary and thermal bounds in turn:



For Step 1, Eqs. (S119) and (S120) give $q_j \le 2x$ and

$$F(2x, x, 1-x) \le \frac{6x^2}{(1-2x)(1-x)^3} < 7x^2, \tag{S125}$$

so $E_{j+1} = 10x^2$ bounds $e_{M_j}(j+1)$. For Step 2, the four factors of A_j are at least $1-x$, $1-x$, $1-6x^2$, and $1-10x^2$. Hence $A_j \ge (1-x)^2(1-16x^2) \ge 1 - 2x - 16x^2$, with $16 = 6 + 10$ and the positive cross terms dropped. For Step 3, since $0 < x \le .01$, thermal squaring yields

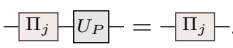
$$f(2x + 16x^2) < 3x^2, \tag{S126}$$

so $\eta_{j+1} = 10x^2$ bounds $1 - P_\tau^o(2M_j, \beta_{j+1})$, and $\varepsilon_{j+1} \le 6x^2 \le 10x^2$ closes the induction at scale $j+1$; the common prefactor 10 covers the three constants 7, 3, and 6.

SV.F. Equal endpoint fields, odd lengths, and the uniform gap

Equation (S124) controls even opposite-field chains at every $j \ge 5$; the theorem needs the thermal purity of odd equal-field chains, which Eq. (S92) turns into the gap. Three things change: the right boundary vector carries the twist U_P, the odd length makes the remainder neither a square nor necessarily positive, and only even lengths are

certified. Define the normalized equal-field partition function $Q_j(n) = Y_n^{\mathrm{s}}(\beta_j)/\lambda_j^n$. The twist is harmless for the leading term, because U_P commutes with the transfer operator and fixes ψ_j by (i), so the overlap c_j survives:



(S127)

For the remainder, pull half of an even length $M \le n$ onto each boundary vector, commuting U_P through $R_j^{M/2}$ ($[U_P, R_j] = 0$ because $[U_P, X_\beta] = 0$ and $U_P\Pi_j = \Pi_j$); the middle operator then has norm at most one ($\|R_j\| \le 1$, U_P orthogonal):

$$
\begin{aligned}
Q_j(n) - c_j &= \bar{v}_j \text{—} R_j^{M/2} \text{—} R_j^{n-M}U_P \text{—} R_j^{M/2} \text{—} v_j \\
&= \bar{y} \text{—} R_j^{n-M}U_P \text{—} y, \qquad y = R_j^{M/2} v_j.
\end{aligned}
\tag{S128}
$$

Cauchy–Schwarz bounds the open contraction by the squared norm of its caps, the even-length opposite-field remainder:

$$
\left| \bar{y} \text{—} \overset{\|\cdot\| \le 1}{R_j^{n-M}U_P} \text{—} y \right| \le
\begin{array}{l} R_j^{M/2} \text{—} v_j \\ R_j^{M/2} \text{—} \bar{v}_j \end{array}
= D_M(j).
\tag{S129}
$$

The odd equal-field chain is thus controlled by the even opposite-field remainder at any even $M \le n$, and the recurrence certifies $M = M_j$.

Fix a scale $j \ge 5$ and an odd length n with $M_j \le n \le 2M_j$. The equal-field purity is a ratio of Q at the two temperatures, as in Eq. (S115), so Q needs an upper bound at β_j and a lower bound at β_{j+1}. Both follow from Eq. (S129) at the certified even length $M = M_j = k_{j+1}$, where the recurrence bounds the remainder at both temperatures:

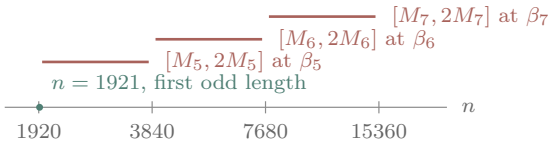
$$
\begin{aligned}
Q_j(n) &\le c_j + D_M(j) = B_M(j), \\
Q_{j+1}(n) &\ge c_{j+1} - D_M(j+1) \\
&= [1 - 2e_M(j+1)]B_M(j+1) \\
&\ge (1 - 2x_{j+1})B_M(j+1) > 0.
\end{aligned}
\tag{S130}
$$

Equation (S115) at length M and Eq. (S130) then bound the equal-field thermal purity factor by factor; the bulk factor is the only term that still depends on n:

$$
\begin{aligned}
P_\tau^{\mathrm{s}}(n, \beta_j) &= r_j^n \frac{Q_{j+1}(n)}{Q_j(n)^2} \\
&\ge \underbrace{(1 - 2x_{j+1})}_{\text{(S130)}} \underbrace{P_\tau^{\mathrm{o}}(M, \beta_j)}_{\ge 1 - x_j} \underbrace{r_j^{n-M}}_{\ge r_j^M,\ n-M \le M} \\
&\ge (1 - 2x_{j+1})(1 - x_j) \underbrace{(1 - \varepsilon_j)(1 - \varepsilon_{j+1})}_{\text{(iii)}} \\
&\ge 1 - 2x_j - 26x_j^2 \\
&\ge 1 - 4x_j,
\end{aligned}
\tag{S131}
$$

where $26 = 20 + 6$ collects $2x_{j+1} = 20x_j^2$ and $\varepsilon_{j+1} \le 6x_j^2$, and $26x_j^2 \le 2x_j$ for $x_j \le 1/13$. Every odd n in $[M_j, 2M_j]$ thus inherits the even-length bound at scale j up to a factor 4, which the proof below converts into the gap.

Proof of Theorem S1. Each odd n gets its own scale j, so the gap bound must be uniform in j. Both $-\log(1 - P_\tau^{\mathrm{s}}(n, \beta_j))$ and the inverse temperature β_j double from one scale to the next, so their ratio in Eq. (S92) is scale independent. The intervals $[M_j, 2M_j]$, $j \ge 5$, cover every odd $n \ge M_5 + 1 = 1921$, each at its own inverse temperature:



Given odd n, choose $j \geq 5$ with $M_j \leq n \leq 2M_j$. Since $x_j \leq x_5 = .01$, Eq. (S131) gives $w_0 \geq P^{\mathrm{s}}_\tau(n, \beta_j) \geq 1 - 4x_j > 1/2$, so H^{s}_n has a unique ground state. At this length n and finite inverse temperature β_j, Eq. (S92) gives

$$\begin{aligned} e^{-\beta_j \Delta_n} &\leq \frac{4x_j}{1-4x_j} && x_j = 10^{-1}\, 10^{-2^{j-5}}, \quad 1 - 4x_j \geq .96, \\ &\leq \frac{5}{12}\, 10^{-2^{j-5}} < 10^{-2^{j-5}}, && \tfrac{.4}{.96} = \tfrac{5}{12} < 1. \end{aligned} \tag{S132}$$

Taking logarithms and dividing by $\beta_j = \beta_5\, 2^{j-5}$ with $\beta_5 = 49 \cdot 2^3 = 392$ proves $\Delta_n > \frac{\log 10}{392}$ for every j. □

In Lean 4, we formalize both the general open-chain theorem from a rational certificate ✓L and the derivation of the four seeds (S105) from the short boundary trace table ✓L, and verify the open-chain integer matrix-product trial contractions on $m = 34, 78, 142, 240$ sites and the nine shortest boundary trace totals directly in the Lean kernel, leaving only the remaining six short-chain trace totals as hypotheses ✓L. Rerunning the same three rational steps of Eq. (S123) from the four seeds (S105) with four-digit rounding and the ring constant 5.583 in place of 6, and entering the quadratic tail at the scale-5 defect 2.241×10^{-4} instead of 10^{-2}, sharpens the uniform bound to $\Delta_n > \log(10^6/2241)/392 \simeq 0.01556$ for every odd $n \geq 1921$ ✓L.